

The role of use for the business value of big data analytics

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ABSTRACT

Big data analytics (BDA) has attracted significant attention from organisations who are seeking to leverage its analytical capabilities to enhance performance. Yet, most BDA investments fail to realise their expected value, indicating that the path for BDA value creation remains unclear; scholars call for a better understanding of the role of BDA use. Therefore, we seek to answer the research question: *How does the use of BDA impact business value?* We conducted a developmental literature review, resulting in the conceptualisation of the BDA-Use-Value (BDA-UV) framework. The framework consists of 10 constructs: BDA system capabilities, BDA use, BDA users and tasks, BDA organisational capabilities, complementary organisational resources, BDA learning, goal-oriented activities, intermediate outcomes, ultimate outcomes, and the internal and external environment. The framework also includes 10 propositions, which illuminate the relationships within the BDA value chain, BDA learning loop, and BDA context. This research contributes to theory by providing a consolidated view of disparate yet commensurate research streams (business value of IT and system use) and is critical in the BDA context where organisations are struggling to leverage BDA's informing potential to derive value.

Introduction

Big data analytics (BDA) holds the potential to improve decision making and enhance firm performance through generating business insights (Dong & Yang, 2020; Grover et al., 2018; Günther et al., 2017). The promise of BDA has seen the global data analytics market rapidly expand, predicted to reach a value of USD 346 billion by 2030 (Market Research Future, 2020). BDA is defined as the techniques, methodologies, practices, applications, systems, and technologies to acquire, store, integrate, and analyse big data (i.e., large structured and unstructured data sets rapidly generated from various sources) (Chen et al., 2012; Mikalef & Krogstie, 2020; Suoniemi et al., 2020). While there have been success stories reporting a positive relationship between BDA and business value¹ (Marr, 2016), “only 20 % of analytic insights will deliver business outcomes” (Trojanos, 2020). This indicates that realising value from BDA is not guaranteed and the path to attaining business value from BDA is unclear (Wang et al., 2019).

To provide insights into the BDA value creation path, scholars have built on two research streams: 1) business value of IT (BVIT); and 2) system use. *BVIT literature* builds largely on the resource-based view of the firm, and explains how IT investments result in organisational performance when supported by organisational complementarities (Melville et al., 2004). *System use literature* –

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¹ For example, Walmart uses BDA to analyse data from various sources (e.g., transactional, social media, economic), enabling them to reduce time spent on identifying a problem and proposing a solution from two weeks to 20 min (Marr, 2016).

progressing from the assumption that technology must be used for benefits to be obtained (Orlikowski, 2000) – explains IT investments result in individual and organisational performance depending on *how* individuals leverage technology to accomplish their goals (Burton-Jones & Grange, 2013). Underpinning both research streams is the rebuttal of technology deterministic assumptions, which posit that IT investments will automatically result in benefits (Huang et al., 2020). However, each stream has contradictory perspectives on the locus of human agency, leading to diverging explanations for *how* IT results in business value. In BVIT literature, agency is located with senior managers recognising their capacity to make strategic decisions and allocate resources to develop organisational complementarities to support IT investments (Kohli & Grover, 2008). In system use literature, agency is located at the level of the individual user and recognises their discretion over whether and how they will use technology when performing their day-to-day tasks (Boudreau & Robey, 2005). These different explanations have resulted in the BVIT literature and system use literature emerging as largely disparate research streams, despite commensurability in their underlying assumption refuting technology determinism and their shared objective of explaining performance outcomes. This has led to a myopic and inadequate understanding of the path to obtaining value from IT investments.

The need to integrate these two streams (i.e., BVIT and system use) has been repeatedly called for (Ruivo et al., 2015; Sabherwal & Jeyaraj, 2015; Schryen, 2013; Soh & Markus, 1995). Furthermore, this integration is of significant contemporary and practical importance in the BDA context, given the necessity for organisations to invest in BDA and their ongoing difficulties of using and benefiting from BDA (Lavalle et al., 2011; Trieu, 2017; Trieu et al., 2022). In literature, BDA value creation is largely approached from a BVIT perspective yielding mixed results, with scholars viewing BDA use – defined as the employment of BDA systems to derive business insights (Trieu, 2017) – as the missing link in the BDA value creation path (Ain et al., 2019; Seddon et al., 2017; Trieu, 2017). Even when BDA use is investigated in conjunction with value, it is often from the perspective of managerial decision makers and overlooks those who analyse data and those who enter and leverage data to make day-to-day decisions at the organisational coalface (Trieu et al., 2022). Such conceptualisation neglects the complex, multi-actor, and multilevel nature of BDA use and its impact on business value. Therefore, our research question is: ‘*How does the use of BDA impact business value?*’.

To answer this question, we develop the BDA-Use-Value (BDA-UV) framework, which illuminates the BDA value creation path and provides an integrated perspective on the broader nomological network surrounding the use and value of BDA. To conceptualise the framework, we perform a developmental literature review (Templier & Paré, 2015), bringing together diverse streams of literature on BDA, system use, and business value from multiple disciplines (i.e., IS as well as business and organisation journals). Multilevel theorising is used to conceptualise the framework because business value is typically examined at an organisational level and system use is typically examined at an individual level. This multilevel perspective provides a comprehensive account of the BDA value creation path and overcomes the incomplete, unrealistic and insufficient view offered by a single level of analysis (Burton-Jones & Gallivan, 2007; Burton-Jones & Volkoff, 2017; Zhang et al., 2019). This framework contributes to theory by providing a consolidated view of different research streams from multiple disciplines, and further contributes to practice by providing insights into how organisations can improve the business value generated from BDA investments, through dedicating purposeful efforts towards both organisational complementarities and the users and use of the BDA.

The remainder of this paper is structured as follows. First, we provide an overview of the focal concepts: BVIT, system use, and BDA. We then describe the developmental literature review method. Subsequently, we conceptualise the multilevel BDA-UV framework. We conclude with a discussion on the novelty and utility of the BDA-UV framework and provide avenues for future research.

Background literature

As our study involves conducting an inductive developmental literature review to construct a conceptual framework, this section serves to provide an initial understanding of the relevant concepts, rather than an explicit theoretical underpinning. Below, we present a brief overview of extant research on the business value of IT (BVIT), system use, and big data analytics (BDA).

Business value of IT

Business value of IT (BVIT) refers to the impacts of IT investments on organisational performance (Kohli & Grover, 2008; Melville et al., 2004). BVIT has received continuous attention in the IS discipline (Melville et al., 2004; Sabherwal & Jeyaraj, 2015), addressing central debates regarding the “productivity paradox” (Brynjolfsson, 1993, p. 67) and the view that “IT doesn’t matter” (Carr, 2003, p. 5). Evidence from research demonstrates IT investments can lead to improved organisational performance when specific conditions – such as complementarities, rarity and inimitability of resources, and competitive barriers – are met (Melville et al., 2004; Soh & Markus, 1995). However, there is a need to continuously advance BVIT knowledge due to ongoing IT advancements in terms of “new tools, platforms and practices” (Sabherwal & Jeyaraj, 2015, p. 816).

One of the most prominent theories used in BVIT research is the ‘*Resource-Based View*’ (RBV) (Fink, 2011; Melville et al., 2004; Rivard et al., 2006; Schryen, 2013). In their seminal review, Melville et al. (2004) argue that RBV is a useful theoretical foundation for examining BVIT resources due to RBV’s focus on resource attributes – such as valuable, rare, imperfectly imitable, and non-substitutability (Barney, 1991) – and the implications of these attributes for efficiency gains and competitive advantage. Schryen (2013) notices that RBV draws attention to the bundling of different IT assets, IT capabilities, and organisational capabilities for value creation and competitive advantage. As such, it is difficult to create value with IT resources alone as value creation also requires complementary organisational resources (Rivard et al., 2006). With RBV as a theoretical foundation, Melville et al. (2004) expand the conceptualisation of BVIT through integrating diverse strands of research. Accordingly, Melville et al. (2004, p. 287) define BVIT as “the organisational performance impacts of IT at both the intermediate process level and the organisation-wide level, comprising both

efficiency impacts and competitive impacts” within three domains (i.e., the focal firm, the competitive environment, and the macro environment). Amongst the factors that shape the relationship between IT investments and organisational performance, Melville et al. (2004) recognise two types of complementarities in the focal firm domain: one between technological and human IT resources (together forming the construct of IT resources); and the other between IT resources and complementary organisational resources (e.g., organisational structure and culture).

Despite RBV being a prominent theory in the development of BVIT models, it has been criticised for being static and is considered insufficient for the study of BVIT in rapidly changing environments (Felt & Gregor, 2014). As such, ‘*Dynamic Capabilities Theory*’ (DCT), which is often seen as an extension of RBV, has become a prominent theory for BVIT research (Daniel & Wilson, 2003; Steininger et al., 2022; Wheeler, 2002). An important insight from DCT for BVIT is that firms need to develop their resource and capability base to achieve and sustain performance outcomes (Steininger et al., 2022). Based on DCT, Wheeler (2002) introduces the Net-Enabled Business Innovation Cycle to understand and predict how emerging technologies can create customer value. Organisational learning features prominently in his framework as it guides the evolution and improvement of the capabilities through human action and as such enables value creation. This important role of organisational learning for value creation with new technologies has also been recognised by other researchers. For example, Daniel and Wilson (2003) emphasise the role of ‘learning-by-doing’ for developing e-business capabilities. Tippins and Sohi (2003) also stress the important role learning plays in enhancing capabilities and achieving performance outcomes with IT. Looking beyond RBV and DCT, research on IT innovation has also recognised the critical role of organisational learning to overcome knowledge barriers that organisations face when assimilating new technologies (Roberts et al., 2017).

Despite the critical need of IT investments, meta-analyses of BVIT research show contradictory results (Kohli & Devaraj, 2003; Sabherwal & Jeyaraj, 2015). To gain insights into these contradictory results, Sabherwal and Jeyaraj (2015) emphasise, amongst other elements, the need to consider IT adoption or use, IT progress, and different value measures when examining the effects of IT investments. This is further substantiated by other scholars who have called for BVIT research to unpack how system use contributes to the realisation of business value (Sabherwal & Jeyaraj, 2015; Soh & Markus, 1995; Trieu, 2017). This requires extending beyond the business process or organisational level, which is currently the predominant focus of BVIT literature, to consider the individual and multi-level nature of system use (Trieu, 2017).

System use

System use – defined as a “user’s employment of one or more features of a system to perform a task” (Burton-Jones & Straub, 2006, p. 231) – has been a central research topic in the Information Systems (IS) discipline for decades (Barkin & Dickson, 1977). System use is conceptualised as comprising three elements: system (the object used), user (the subject using a system), and task (a goal-directed activity) (Burton-Jones & Straub, 2006).

Several perspectives on system use have been investigated with increasing sophistication and theorisation over time (Burton-Jones et al., 2017). Early research stressed the importance and complexity of system use (Srinivasan, 1985). Research then progressed to examining system use as the behavioural actions and doings of individuals (Compeau & Higgins, 1995). This was followed by IT acceptance research which conceptualises system use as a dependent variable (Davis, 1985; Venkatesh et al., 2016). And then came the concept of *post-adoptive system use*, which “examines IS use as part of an ongoing process with the aim of understanding how it is shaped by and in turn shapes a variety of other phenomena at multiple levels of analysis” (Burton-Jones et al., 2017). In this study, we focus on post-adoptive system use because, while system use is a necessary condition for realising benefits from technologies, it is also vital to understand what elements influence system use and subsequently influence value creation.

Many conceptualisations of post-adoptive system use have emerged, varying in whether users accept or resist a system and whether or not their use of a system conforms to the organisational intent (Beaudry et al., 2020). Recognising these distinctions, Beaudry et al. (2020) developed a taxonomy consisting of four types of system use: conforming acceptance, conforming resistance, non-conforming acceptance, and non-conforming resistance. The *conforming acceptance* type of system use is reflective of users both embracing the system and using the system in a way that conforms with the organisation’s intent. This type of system use has been found to favourably mediate the relationship between systems and individual and organisational performance (Beaudry et al., 2020). This contrasts with resistance and non-conforming system use behaviours, which are largely associated with detrimental impacts (Beaudry et al., 2020), such as inefficient workflows and impaired productivity (e.g., Kane & Labianca, 2011).

As research into the conforming-acceptance type of system use has progressed, richer conceptualisations have emerged focusing on robustly theorising how individuals use systems to attain individual and organisational goals (Trieu et al., 2022). For example, effective use examines how individuals transparently interact with systems to extract faithful information that they then leverage to make informed decisions in support of the accomplishment of their goals, which improves their performance (Burton-Jones & Grange, 2013). To facilitate effective use, individuals can engage in other forms of the conformance-acceptance type of system use – such as adaptive use and innovative use – by revising their use of features and “discovering new ways” of using the system to perform their tasks (Sun et al., 2019, p. 401).

In addition to post-adoptive system use, research has stressed the need to study system use as a *multilevel* phenomenon (Burton-Jones & Gallivan, 2007; Zhang et al., 2019). For instance, Burton-Jones and Gallivan (2007) provide guidelines to study the function, structure, and context of use when developing a multilevel theory of system use. Likewise, Zhang et al. (2019) highlight the importance of clarifying the levels of the construct and specifying the types of relationships among constructs (i.e., causal and non-causal, and single-level and cross-level) when conducting multilevel research. Despite these guidelines, a limited number of studies have been conducted accordingly, raising the concern of “an unnatural, incomplete and disjointed view of how information systems are used in

practice” (Burton-Jones & Gallivan, 2007, p. 657).

Overall, the challenges apparent in both system use and BVIT literatures surrounding the necessity of incorporating richer and multilevel conceptualisations of system use to better explain the value creation path is also evident in the BDA context.

Big data analytics

Due to technological advancements, increasing numbers of machines, devices, and people are connected to digital networks, changing the way data is created, communicated, shared, accessed, integrated, and used (Grover et al., 2018). This phenomenon, known as *big data*, refers to “the large magnitude of data” that is rapidly produced from various sources and may contain noise, uncertainty, and errors (Grover et al., 2018, p. 389), and as such, is characterised by “volume, velocity, variety and veracity” respectively (Surbakti et al., 2020, p. 1). Because big data is “a shifting and evolving cultural and scholarly phenomenon”, it is a dynamic concept that can be used in different contexts in heterogeneous ways (Favaretto et al., 2020, p. 17). It may need to be deconstructed into different constituents to more correctly capture its “essence and characteristics” and consequently its application in big data analytics (BDA) (Favaretto et al., 2020, p. 17).

BDA inherits the unique characteristics of big data and evolves from business intelligence (BI) and business analytics (BA) (Dong & Yang, 2020; Grover et al., 2018; Trieu, 2017). From the 1990 s, BI systems have been used for measuring performance and justifying actions related to targets (Lavalle et al., 2011). Evolving from BI, BA emerged in the early 2000s enabling organisations to use analytics to discover insights to guide their actions (Krishnamoorthi & Mathew, 2018). Evolving from BA, BDA is designed for processing big data from various sources to prescribe actions and generate insights for data consumers to make decisions (Lavalle et al., 2011; Seddon et al., 2017). Understanding how data and information reveal insights leading to strategic, tactical, and operational decision-making is critical (Rothberg & Erickson, 2017).

Extant research has produced inconsistent and contradictory findings regarding the impact of BDA on organisational performance (e.g., Akter et al., 2016; Božić & Dimovski, 2019; Ghasemaghahi, 2019a). A meta-analysis by Oesterreich et al. (2022) shows that these contradictory findings stem from many factors including methodological and conceptual oversights regarding value creation mechanisms, lack of consideration of contextual factors, and variability in the appropriateness of the independent and dependent variables included in the studies. Consequently, Oesterreich et al. (2022, p. 11) stress that BDA should be considered as “a sociotechnical system consisting of technical and social components” because technical factors (e.g., “BDA technical resources and data quality”) are necessary but insufficient to contribute to organisational performance. Rather, social factors (e.g., “human resources, management capabilities, and organisational culture”) are also key to attaining desired results (Oesterreich et al., 2022, p. 11). Mikalef et al. (2018) further argue that the mechanisms behind how BDA leads to competitive advantage remain unclear. As such, there is a need to identify the mechanisms underpinning how BDA insights can be turned into actions, which contribute to competitive advantage. The effective use of BDA is proffered to be the key mechanism to explain the mixed results in past studies (Ain et al., 2019; Trieu, 2017). Yet, richer and multilevel conceptualisations are needed of the specific types of BDA use that are associated with desired performance outcomes (Trieu, 2017; Trieu et al., 2022).

Summary of background literature

From a BVIT perspective, diverse studies have been conducted with the inclusion of various factors to explain how organisations obtain outcomes from their IT investments (Melville et al., 2004; Sabherwal & Jeyaraj, 2015). Amongst these factors, system use has received limited attention but should be investigated further; as expressed by Soh and Markus (1995, p. 30), “IT cannot improve organizational performance unless the technology is used”. From a system use perspective, post-adoptive use is a prominent but complex concept that plays a critical role in further understanding BDA value creation. However, there are several important gaps. Although some studies respond to the call to investigate different types of system use (Burton-Jones et al., 2017), research examining how different types of system use ultimately contribute to organisation performance is lacking (Beaudry et al., 2020). Furthermore, studies from both system use and BVIT research suggest examining the phenomenon from a multilevel perspective to gain a more complete view of system use in practice. Yet, research conducted in such a manner remains scarce. From a BDA perspective, extant research highlights the importance of investigating users’ roles and what elements influence users interacting with BDA, which is in line with the gaps identified from studies on BVIT and system use. As such, exploring the relationship among BDA, business value, and system use from a multilevel perspective is essential to uncover the path from BDA investments to positive impacts.

Research methodology

We performed a developmental literature review to consolidate research at the intersection of BDA, system use, and business value. A developmental literature review allows the incorporation of conceptual and empirical studies focusing on “theoretically dissimilar concepts and findings” – in this case system use and BVIT which have emerged as separate streams of research – to develop a conceptual framework (Templier & Paré, 2015, p. 118). We adhered to Templier and Paré’s (2015) guidelines for conducting literature reviews, which consist of six steps: 1) formulating the problem; 2) searching the literature; 3) screening for inclusion; 4) assessing quality; 5) extracting data; and 6) analysing and synthesising data.

Step 1 & Step 2: Formulating the problem & searching the literature

As mentioned in the Introduction, the problem this review is seeking to address (Step 1) is the need to provide robust insights into the link between BDA, system use, and business value. To search the literature (Step 2), we examined both IS and business journals. For the IS journals, we began with the AIS Senior Scholars' List of Premiere Journals² as it provides a robust and rich foundation of high-quality outputs. To improve coverage, consistent with Morton et al. (2022), we complemented our search with additional IS journals.³ To identify relevant business journals, consistent with Oehlhorn et al. (2020), we leveraged Lowry et al. (2004) and narrowed the search to the top business and organisational journals for IS researchers.⁴ The journals were searched using synonyms of BDA, system use, and business value. Accordingly, the search statement was: ("big data" OR "big data analytics" OR "business analytics" OR "business intelligence") AND ("business value" OR "strategic value" OR "IT value" OR "competitive advantage" OR "performance" OR "benefit") AND ("use" OR "usage"). The search was not restricted by time and included all articles published up to the end of 2023. We used EndNote as our reference management software to manage articles. In total, 1816 papers were selected for initial screening.

Step 3: Screening for inclusion

These papers were divided amongst three researchers to screen for inclusion (Step 3). Prior to screening, the inclusion and exclusion criteria were formulated. Articles were included if they examined at least two of the three concepts (i.e., BDA, system use, value) central to our research question. Articles were excluded if: 1) they were not a research article (e.g., discussion notes, tutorials, editorials); 2) they primarily focused on technical aspects related to prototype or algorithm development; or 3) when examining system use, they did so from the perspective of adoption or implementation rather than examining use as the behaviour of interacting with a system. To assess eligibility, three rounds of screening were performed to evaluate the relevancy of retrieved papers (Fig. 1).

In the first round, 117 papers were excluded because they were inappropriate paper types (e.g., discussion notes, editorials), with 1699 papers remaining. In the second round, we read the abstract, title, and keywords of the remaining articles, and 1363 articles were excluded as two of the three core concepts (BDA, system use, and business value) were not prominent. The high number of excluded articles was expected given that some of the search terms (e.g., use, performance, value) also have a generic meaning and were used in a way that were not relevant to the research problem. This resulted in 336 articles remaining. In the last round, we read the papers in full, resulting in 213 articles being excluded as they focused predominantly on only one of the three main concepts. Throughout the screening process, to ensure consistent application of the inclusion/exclusion criteria, the researchers engaged in multiple discussions with any uncertainties raised and deliberated collectively. Furthermore, during the second round of screening, whenever it was unclear an abstract should be included or excluded, it was deferred for full text review and further discussion. Ultimately, 123 articles were identified as relevant (Appendix 1).

Step 4: Assessing quality

According to Templier and Paré (2015), quality assessments (Step 4) are not required for developmental reviews. However, according to Webster and Watson (2002), quality appraisals are important and begin with searching the literature. As such, our search was narrowed to high quality IS, business, and organisation journals. Such an approach is in line with developmental reviews as an exhaustive set of papers is not required; rather, only a sample which covers core concepts is needed (Templier & Paré, 2015).

Step 5 & Step 6: Extracting data & analysing and synthesising data

The extraction (Step 5) and analysis and synthesis of data (Step 6) were manually performed in two stages using NVivo (QSR International, v.14). The first stage involved the inductive analysis of papers from the original AIS basket of eight journals – as well as the journal Decision Support Systems because of its focus on BDA. This stage led to the development of the constructs, relationships, and multilevel theorising present in the BDA-UV framework. The second stage involved the deductive analysis of papers from the remaining IS journals and business and organisational journals, with the BDA-UV framework used as a deductive classification framework. During the deductive analysis, we were open to new constructs and relationships emerging, which were then inductively analysed and used to refine the BDA-UV framework.

Both stages (i.e., inductive and deductive analysis) were independently performed by the first author. Throughout both stages, actions were taken to ensure the reliability and credibility of the data extraction, analysis, and synthesis. This included the ongoing development and refinement of coding rules and a code book to ensure the analysis was consistent (Bandara & Syed, 2023). In addition, multiple discussion sessions were held with researchers with expertise in BDA, system use, and BVIT to review all codes and

² European Journal of Information Systems, Information Systems Journal, Information Systems Research, Journal of the Association for Information Systems, Journal of Information Technology, Journal of Management Information Systems, Journal of Strategic Information Systems, Management Information Systems Quarterly, Decision Support Systems, Information and Organization, and Information and Management.

³ Communications of the Association for Information Systems, and Information Systems Frontiers.

⁴ Academy of Management Journal, Academy of Management Review, Administrative Sciences Quarterly, Harvard Business Review, Journal of Marketing, Management Science, MIT Sloan Management Review, Organization Science, Organizational Behaviour and Human Decision Processes, Strategic Management Journal.

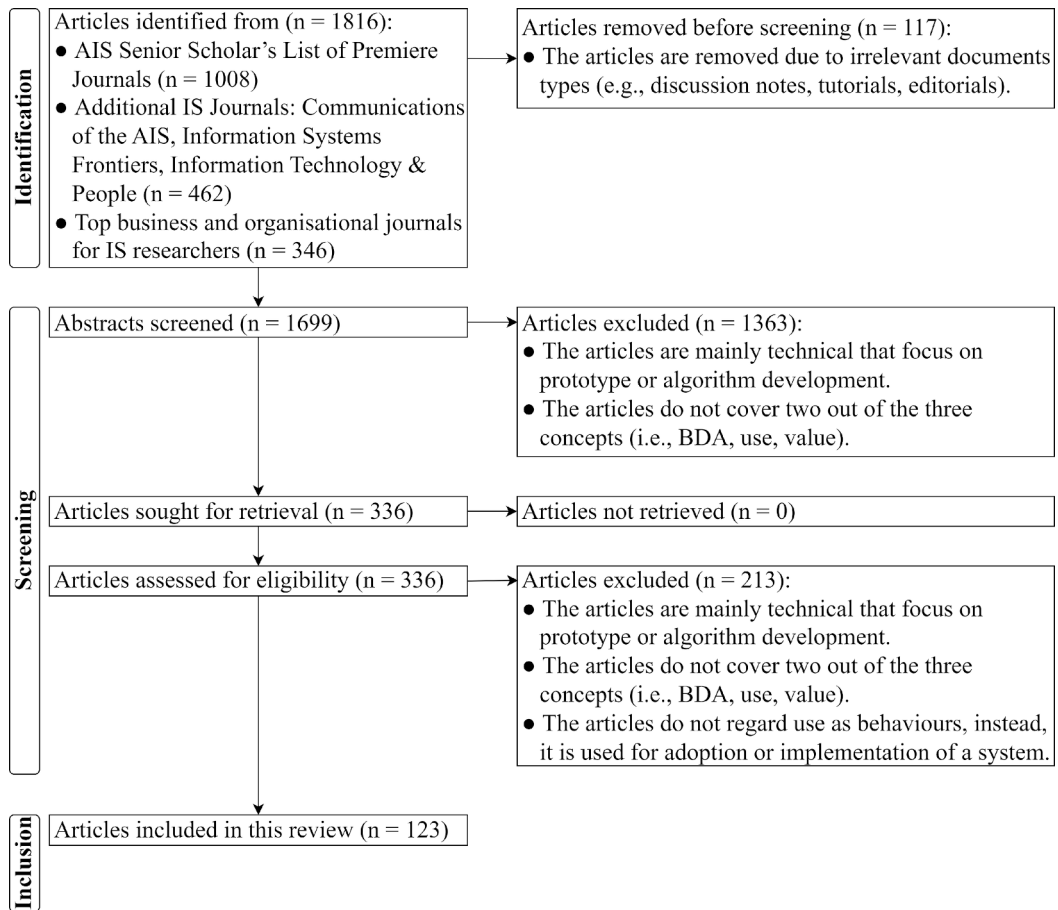


Fig. 1. The Process of Selecting Articles for Inclusion (Step 3).

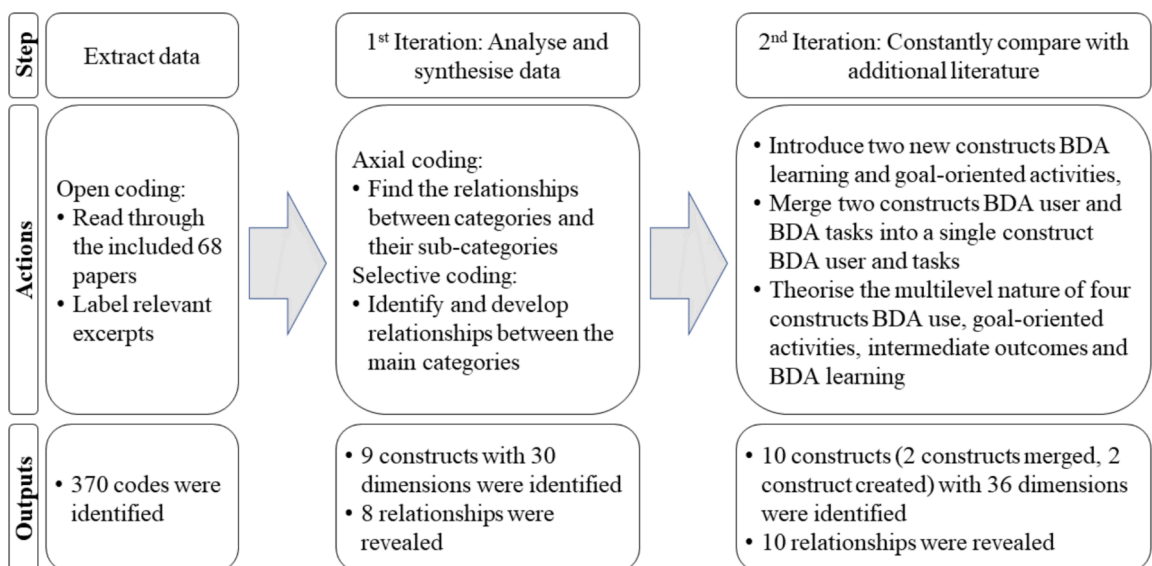


Fig. 2. The Process of Inductively Extracting, Analysing, and Synthesising Data (Steps 5 & 6).

resolve uncertainties and discrepancies (Saldaña, 2021). Below, we discuss the extraction and analysis involved in Stage 1, followed by Stage 2.

Stage 1: Inductive data extraction and analysis

The inductive data extraction and analysis process (Fig. 2) was guided by Wolfswinkel et al. (2013). To extract data (Step 5), we performed open coding by reading through each paper and labelling relevant excerpts. We opted for open coding because “developmental reviews are particularly valued for proposing new theoretical foundations, for developing new approaches to address existing research problems” (Templier & Paré, 2015, p. 119). Therefore, a deductive classification framework that relied on existing theoretical explanations for BVIT and system use was not developed prior to open coding. Upon completing open coding, 370 codes were extracted. To analyse and synthesise data (Step 6), two iterations were performed, as elaborated below. We used axial and selective coding (Iteration 1) and constant comparison with broader literature (Iteration 2) to identify and refine the constructs and relationships.

First iteration. Axial and selective coding were used in the first iteration of data analysis as it provides a disciplined approach to analysing and consolidating findings (Templier & Paré, 2015). In accordance with Wolfswinkel et al. (2013, p. 7), axial coding “is about finding the relationships between categories and their sub categories, selective coding is the process of identifying and developing relations between the main categories”. After the first iteration, nine constructs with their dimensions were identified: *BDA system*, *BDA use*, *BDA organisational capabilities*, *complementary organisational resources*, *user characteristics*, *task characteristics*, *intermediate outcomes*, *ultimate outcomes*, and *internal and external environment*. Initial selective coding revealed relationships between these constructs, whereby *BDA systems* – in conjunction with *BDA organisational capabilities* and *complementary organisational resources* (from the BVIT perspective), as well as *user and task characteristics* (from the system use perspective) – influence *BDA use*. Subsequently, through *BDA use*, *intermediate outcomes* will result, which accumulate over time to generate *ultimate outcomes*.

Second iteration. Following our initial axial and selective coding, we constantly compared our analyses to literature, which revealed the need to unpack the multilevel nature of *BDA use* and business value. Consequently, a second iteration of the analysis was performed, which involved incorporating multilevel theory building literature. This resulted in three core amendments:

- 1) The introduction of two new constructs: *BDA learning* (previously part of *BDA organisational capabilities*) and *goal-oriented activities* (previously part of *BDA use* and *intermediate outcomes*). The inclusion of *BDA learning* recognised the potential for feedback loops for learning to be initiated after *BDA use* (Barki et al., 2007; Burton-Jones & Grange, 2013; Jasperson et al., 2005). The inclusion of *goal-oriented activities* recognised that, prior to yielding an *intermediate outcome* in the BDA context, users need to not only use BDA but also perform other activities that are goal-oriented (Seddon et al., 2017; Strong et al., 2014).
- 2) The merging of two constructs, *BDA users* and *BDA tasks*, into a single construct *BDA users and tasks* to recognise the goal-oriented nature of users in completing their tasks. This resulted in users being categorised as: data collectors who perform data collection tasks; data analysts who perform data analysis tasks; and data consumers who perform data consumption tasks (Murphy et al., 2012).
- 3) Theorising the multilevel nature of four constructs: *BDA use*, *goal-oriented activities*, *intermediate outcomes*, and *BDA learning*. This involved identifying: i) the individual, collective, and organisational levels for each multilevel construct; ii) the hierarchical relationships involved where a lower level enacts a higher level, and the higher level assigns and enacts the lower level (Burton-Jones & Gallivan, 2007); and, iii) the nature of relationships between constructs (e.g., single level/cross level, causal/non-causal (i.e., functional equivalence)) (Zhang et al., 2019).

Overall, ten constructs with 36 sub-categories (Appendix 2) and ten propositions were identified.

Stage 2: Deductive data extraction and analysis

The second stage involved the data extraction and analysis of the relevant articles from the remaining IS, business and organisational journals. The constructs (Appendix 2) and propositions in the BDA-UV framework served as a deductive classification framework to aid data extraction (Step 5) and improve the consistency of the coding (Templier & Paré, 2015). During this process, we were open to emerging insights which were inductively analysed (Bingham, 2023). Overall, the data extracted was consistent with the constructs and propositions within the BDA-UV framework. Within the constructs, additional specificity was identified with a new dimension under *complementary organisational resources* (i.e., non-technological resources) and additional examples under the dimensions of *BDA users and tasks*, *BDA organisational capabilities*, and *ultimate outcomes*. As the overall framework has remained consistent, this provides confidence that theoretical saturation has been reached.

For the analysis and synthesis (Step 6), an initial comparison was performed to identify if there were any core differences between the relevant literature extracted from IS journals versus the business and organisational journals. This comparison demonstrated that the business and organisational journals were largely consistent with IS journals, although the IS journals provided richer conceptualisations of the focal constructs. Due to the consistency across the stages of the analysis and the disciplines included in the analysis, the insights from Stage 1 and Stage 2 were consolidated to justify the propositions present within the BDA-UV framework.

BDA-UV multilevel conceptual framework

The multilevel BDA-UV framework illustrated in Fig. 3 includes ten relationships between ten constructs: *BDA system capabilities*, *BDA users and tasks*, *BDA use*, *goal-oriented activities*, *intermediate outcomes*, *ultimate outcomes*, *BDA organisational capabilities*,

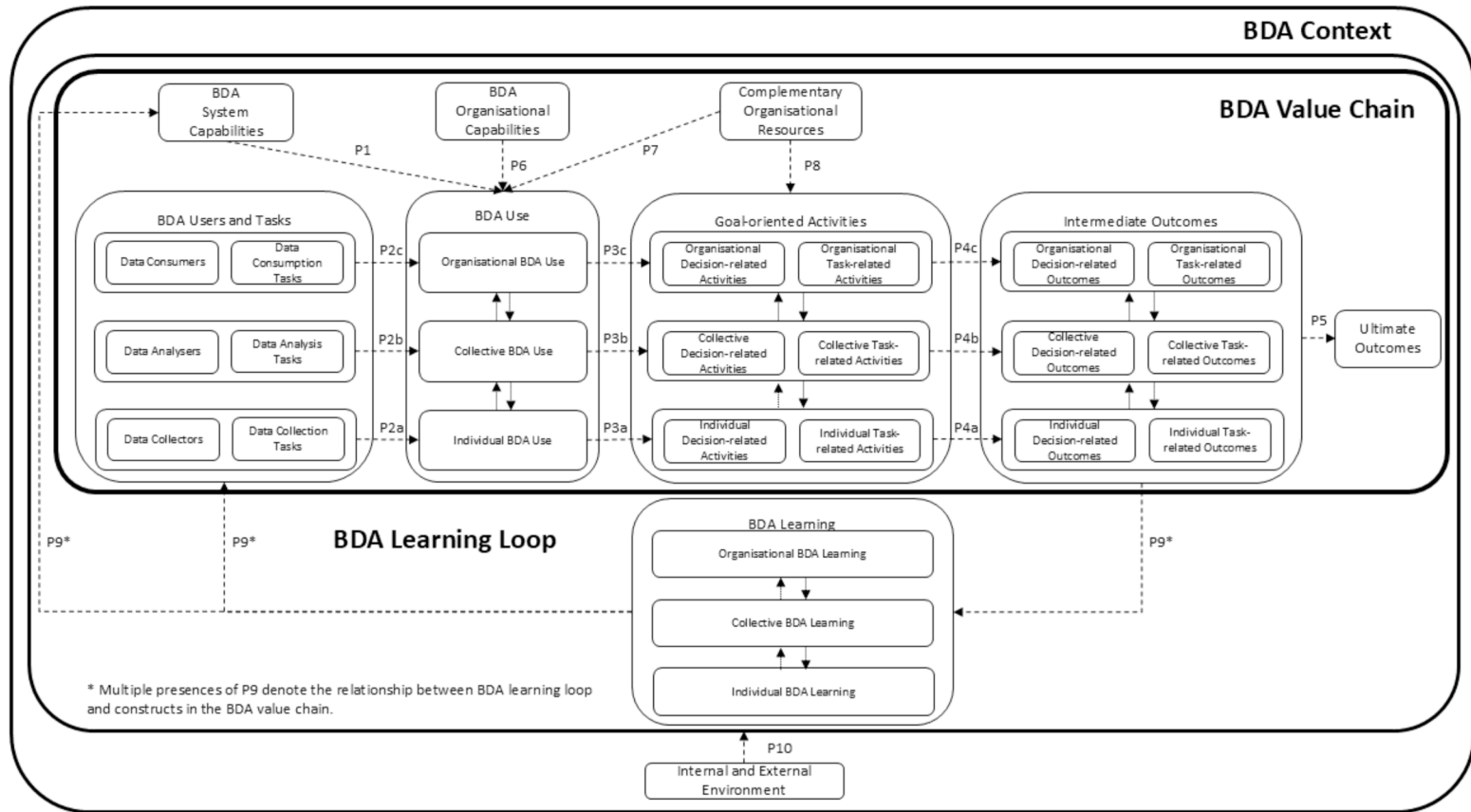


Fig. 3. BDA-UV Framework.

complementary organisational resources, BDA learning, and internal and external environment. We identify three categories of relationships – including the BDA value chain (P1 to P8), BDA learning loop (P9), and a contextual factor moderating the relationships (P10) – with supporting evidence in Table 3. The relationships in Fig. 3 are denoted with a dashed line as we are not drawing causal inferences (consistent with Tate et al., 2015). Below, we first introduce the constructs and then propose their relationships from a multilevel perspective.

BDA-UV constructs

The BDA-UV constructs are described in Table 1 and the dimensions and examples underpinning each construct are outlined in Appendix 2. From the system use literature, we identify six constructs: four operate mainly at the individual level (*BDA users and tasks*, *BDA use*, *goal-oriented activities*, and *BDA learning*); one mainly operates at the system level (*BDA system capabilities*); and another typically operates across multiple levels (*intermediate outcomes*). From the BVIT literature, we identify eight constructs: six operate mainly at the organisational level (*BDA use*, *intermediate outcomes*, *ultimate outcomes*, *BDA organisational capabilities*, *complementary organisational resources*, and *BDA learning*); one mainly operates across multiple levels (*goal-oriented activities*); and one is a contextual factor (*internal and external environment*). Since the construct of *BDA use* operates at both individual and organisational levels, we see its prominent role in bridging the literatures on system use and business value in the context of BDA. Below, we provide more insights into the multilevel constructs in the BDA-UV framework, with examples provided in Table 2.

BDA use is a multilevel construct involving individual, collective, and organisational levels (see Table 1 for definition, Fig. 3 for role/context, and Table 2 for its multilevel nature). All levels of *BDA use* are conceptually similar, originating from identical behavioural processes (Zhang et al., 2019), and aligning broadly with the definition of *BDA use*: users' employment of a BDA system's features (direct use) or outputs (indirect use) to perform goal-oriented activities. *Individual BDA use* refers to an individual user using a BDA system's features or its outputs to perform individual goal-oriented activities. *Collective BDA use* refers to interdependent users with shared goals collectively using a BDA system's features or outputs to perform collective goal-oriented activities. *Organisational BDA use* refers to an institution consisting of interdependent individuals and collectives with organisational goals using a BDA system's features or outputs to perform organisational goal-oriented activities, such as supporting activities across functional areas (Wang et al., 2019). Structural relationships exist between these levels whereby the actions and interactions among individuals at lower levels (individual/collective level) generate use at higher levels (collective/organisational level) (Burton-Jones & Gallivan, 2007; Zhang et al., 2019).

Goal-oriented activities (see Table 1 for definition, Fig. 3 for role/context, and Table 2 for its multilevel nature) reflects the decision- and task-related activities that need to be performed to complete users' tasks. The construct of *goal-oriented activities* aligns with

Table 1
Identified Constructs: Description, Sources, Levels.

Construct	Description	Reference	Source	Level
BDA System Capabilities	The BDA assets, the features and affordances, the quality of the BDA system and the data contained within, and the strategic characteristics of big data and the technologies that process it.	Chen et al. (2012)	System use & BVIT	System
BDA Users and Tasks	An individual who directly or indirectly uses a BDA system in a role of a data collector, data analyser, or data consumer to perform his/her goal-directed activities (i.e., data collection, analysis, or consumption tasks).	Burton-Jones and Straub (2006); Murphy et al. (2012)	System use	Individual
BDA Use	Users' employment of a BDA system's features (direct use) or outputs (indirect use) to perform goal-oriented activities.	Burton-Jones and Straub (2006)	System use & BVIT	Organisational *Collective Individual
Goal-oriented Activities	Users' actions upon the outputs they obtain from using a BDA system that accomplish intermediate outcomes.	Burton-Jones and Grange (2013); Seddon et al. (2017)	System use & BVIT	Organisational Collective Individual
Intermediate Outcomes	Relatively short-term, immediate benefits obtained from goal-oriented activities.	Burton-Jones and Volkoff (2017); Seddon et al. (2017)	System use & BVIT	Organisational Collective Individual
Ultimate Outcomes	Relatively long-term business value contributing to organisational performance accumulated from intermediate outcomes.	Seddon et al. (2017)	System use & BVIT	Organisational
BDA Organisational Capabilities	Organisational BDA-specific abilities to effectively implement and adopt BDA systems towards the generation of ultimate outcomes.	Pappas et al. (2018)	BVIT	Organisational
Complementary Organisational Resources	Organisational non-BDA resources which have synergies with BDA systems.	Melville et al. (2004)	BVIT	Organisational
BDA Learning	Actions taken to improve BDA use and elements influencing BDA use.	Burton-Jones and Grange (2013); Grover et al. (2018)	System use & BVIT	Organisational *Collective Individual
Internal and External Environment	Internal (micro/organisational environment) and external (meso and macro/competitive environment) surroundings of or influences on BDA use, goal-oriented activities, intermediate outcomes, or ultimate outcomes.	Melville et al. (2004); Mikalef and Krogtie (2020)	BVIT	Meso / Macro

* Informed by multilevel theorising and not directly observed from articles.

Table 2
Multilevel Constructs: Definitions, Examples, and Descriptions.

Construct Overarching Definition	Individual Level		Collective Level		Organisational Level	
	Description	Example	Description	Example	Description	Example
BDA Use Users' employment of a BDA system's features (direct use) or its outputs (indirect use) to perform goal-oriented activities.	An individual user employing a BDA system's features (direct use) or its outputs (indirect use) to perform individual goal-oriented activities.	Nurses as data collectors perform individual-level use when they enter patient data during a patient visit.	Interdependent users with shared goals collectively utilising a BDA system's features or outputs to perform collective goal-oriented activities.	A health informatics team as a collective of data analysers performs data analysis and develops dashboards through using data entered by data collectors along with other data automatically generated by the system (e.g., patient number).	An institution consisting of interdependent individuals and collectives with organisational goals utilising a BDA system's features or outputs to perform organisational goal-oriented activities.	Data consumers, such as department supervisors, access the dashboard that presents the analysis result prepared by the data analysers.
Goal-oriented Activities Users' actions upon the outputs they obtain from using a BDA system that accomplish intermediate outcomes.	An individual user making decisions and performing tasks based on outputs of using a BDA system that intend to create individual intermediate outcomes.	Clinicians decide the patient's treatment and treat the patient after accessing a patient's past medical records and enter patient's details.	Interdependent users' interaction for making decisions and performing tasks based on outputs of using a BDA system that intend to create collective intermediate outcomes.	Data analysers form an interdisciplinary team of clinicians and health informatics personnel analyse aggregate data to provide a care plan for complex patients.	The institution making decisions and performing tasks based on outputs of using a BDA system that intend to create organisational intermediate outcomes.	The supervisors make managerial decisions and allocate resources based on the analysis result and proposed care plan.
Intermediate Outcomes Relatively short-term, immediate benefits delivered from goal-oriented activities.	Individual decision and task benefits obtained from individual BDA use and goal-oriented activities.	Accurately recording required data enables nurses to make appropriate operational decisions.	Collective decision and task benefits obtained from collective BDA use and goal-oriented activities and aggregated from individual benefits.	The analysis result enables the clinical team to make more actionable decisions and provide a better care plan for the patient.	Organisational decision and task benefits obtained from organisational BDA use and goal-oriented activities and aggregated from individual and/or collective benefits.	The result along with the complete care plan enable the supervisors to improve the quality of decision-making and allocation of resources.
BDA Learning Actions taken to improve BDA use, its antecedents (e.g., BDA system capabilities, BDA users tasks) and BDA organisational capabilities.	An individual behaviour to develop interpretations and understanding from existing and new information regarding BDA system capabilities and/or BDA use.	When a nurse experiences low task efficiency due to difficulties with using the system's features, it may initiate learning behaviours.	A process a team interacts with team members and acts upon feedback regarding BDA system capabilities and/or BDA use to adapt or improve.	The nurse communicates with colleagues who share similar experience or have a better understanding of the system to collectively learn how to utilise the system features.	A process to institutionalise and legitimate common understanding and beliefs regarding BDA system capabilities and/or BDA use.	A nurse supervisor recognises it is necessary to provide information to enhance the understanding of the system, and thus, this information will be institutionalised.

'informed action', which is a dimension of effective use. Informed action is defined as "the extent to which a user acts upon the faithful representations he or she obtains from the system to improve his or her state" (Burton-Jones & Grange, 2013, p. 642). We segment *goal-oriented activities* from *BDA use* because it is necessary to highlight the nature of BDA users acting on representations from BDA systems to make decisions and complete tasks. This is consistent with Seddon et al. (2017) who indicate that subsequent to using BDA to generate insights, decisions need to be made which will lead to intentional value-creating actions. We merge decision-making and these actions because they are goal-oriented and occur after *BDA use* but prior to obtaining *intermediate outcomes*. We consider *goal-oriented activities* to operate at multiple levels because individuals can pursue different goals from collectives and their organisation (Nevo et al., 2021). *Goal-oriented activities* at different levels share a conceptual meaning and originate from identical behavioural processes (Zhang et al., 2019): users acting upon the outputs obtained from using a BDA system to make and execute decisions.

Intermediate outcomes (see Table 1 for definition, Fig. 3 for role/context, and Table 2 for its multilevel nature) is a multilevel construct operating at individual, collective, and organisational levels that represents the immediate, short-term decision- and task-related outcomes gained from *goal-oriented activities* following *BDA use*. Following the same reasoning as *goal-oriented activities*, improved decision-making (decision-related outcomes) is necessary but insufficient to improve task performance (task-related outcomes). The task also needs to be executed properly to lead improvements. Consistent with the construct 'immediate concrete outcome' in Strong et al. (2014), *intermediate outcomes* is a specific outcome generated from taking actions to attain short-term

individual or organisational goals that collectively contribute to long-term organisational goals.

In the system use literature, learning occurs at an individual level and is performed to improve system use (Burton-Jones & Grange, 2013). Conversely, in BVIT literature, learning occurs at an organisational level and is associated with improving BDA system and organisational capabilities (Grover et al., 2018). Although most research uses a single-level perspective to study learning, we conceive *BDA learning* as a multilevel phenomenon consistent with Wiewiora et al. (2019). Accordingly, *BDA learning* (see Table 1 for definition, Fig. 3 for role/context, and Table 2 for its multilevel nature) is conceptualised as a multilevel construct spanning individual, collective, and organisational levels. It involves the actions taken to improve *BDA use* by changing cognition and actions regarding *BDA system capabilities* and *BDA use* of individual and collectives embedded in an organisation (Wiewiora et al., 2019). *BDA learning* can occur top-down when organisations exploit and make existing knowledge available to individuals and collectives, and bottom-up when individuals and collectives explore new knowledge that becomes institutionalised at the organisational level (Crossan et al., 1999).

BDA-UV construct relationships

The BDA-UV framework, consists of 10 constructs and 10 propositions organised into three themes (Fig. 3): the BDA value chain (P1 to P8), the BDA learning loop (P9), and the BDA contextual environment (P10). As these relationships are situated in a multilevel framework, two types of relationships are apparent: cross-level and functionally equivalent (per Zhang et al., 2019). A *cross-level relationship* (CL) refers to “when the origin and the target belong to a different level” (Zhang et al., 2019, p. 11). This is captured in P1 (system-level to multilevel), P2 (individual-level to multilevel), P5 (multilevel to organisational-level), P6, P7, P8 (organisational-level to multilevel), the second process of P9 (i.e., multi-level (*BDA learning*) to individual-level (*BDA users and tasks*) and system-level (*BDA system capabilities*)). A *functionally equivalent relationship* (FE) describes a construct that has the same effects at multiple levels (Burton-Jones & Gallivan, 2007; Zhang et al., 2019). This is captured in P3, P4, and the first process of P9 (i.e., multilevel (*intermediate outcomes*) to multilevel (*BDA learning*)). For example, individual, collective, and organisational *BDA use* influences individual, collective, and organisational *goal-oriented activities* respectively.

The following subsections explain each of the three categories – BDA value chain, BDA learning loop, and context – and derive their related propositions. Table 3 lists all ten propositions and the supporting evidence for each.

BDA value chain

The BDA value chain describes how BDA system capabilities are used by BDA users to perform goal-oriented activities to complete their tasks and obtain intermediate outcomes, accumulating to ultimate outcomes. Below, we propose the BDA value chain relationships.

BDA system capabilities consist of the BDA assets, the features and affordances, the quality of the BDA system and the data contained within, and the strategic characteristics of big data and the technologies that process it. A variety of BDA assets have been explored – including the technology (e.g., “complex data and analytics technologies” (Chae, 2014, p. 2), “model-oriented and data-oriented decision support systems” (Trieu, 2017, p. 111), “data warehouses, online analytical processing (OLAP), statistical and quantitative tools, visualisation tools and data mining tools” (Seddon et al., 2017, p. 237)), its operating infrastructure (Kettinger et al., 2013), and underlying data (e.g., “large structured and unstructured data sets” (Mikalef & Krogstie, 2020, p. 267)). Consistent with extant research on system use (e.g., Burton-Jones & Straub, 2006), these BDA assets need to be implemented before users can use BDA. When examining BDA assets, some scholars have focused on their material features and the affordances that arise from the “potentials for actions that arise from the relationship between technical objects (e.g., the materiality of BDA technologies) and goal-oriented users or groups of users” (Lehrer et al., 2018, p. 430). These affordances are then actualised during the use of the system (Burton-Jones & Volkoff, 2017; Lehrer et al., 2018).

The quality of the BDA assets and features (e.g., system quality: flexible, integrated, versatile; data quality: accurate, comprehensive, correct, consistent (Torres et al., 2018)) represent a necessary condition for the use of BDA (Trieu, 2017). According to Ghasemaghaei and Turel (2022, p. 486), “with the increased variety, velocity and volume of... big data”, the complexity of BDA systems has increased, placing more cognitive demands on users. This has resulted in users performing workarounds and using BDA in noncompliant ways to complete their tasks and to “bypass, overcome or reduce the impact of obstacles that are perceived as preventing individuals from achieving desired goals” (Ghasemaghaei & Turel, 2022, p. 485). BVIT literature that has drawn on RBV has identified that when BDA is valuable, rare, costly to imitate, and organisationally embedded, it is more likely to be used in a way to generate benefits (Grover et al., 2018).

This research proposes that, taken together, *BDA system capabilities* influence *BDA use*. As specified in the ‘BDA-UV Constructs’ section, *BDA system capabilities* operates at a system level, whereas *BDA use* operates at multiple levels. As such, the relationship is theorised to be cross-level, indicating the capabilities can be used individually by a single user, collectively by a group of interdependent and goal-oriented users, or organisationally by an institution consisting of interdependent individuals and collectives. Thus, it is proposed:

Proposition 1. *BDA system capabilities influence BDA use.*

Users and tasks are critical elements of system use (Burton-Jones & Straub, 2006). Due to the roles BDA users perform in relation to their tasks, we merge them into the construct *BDA users and tasks* and recognise three user roles. The first role is data collectors who perform data collections tasks, such as inputting and accessing data on a daily basis to achieve their operational goals. These users are focused on their immediate tasks and have a limited understanding of “how data is used outside their own occupational boundary” (Murphy et al., 2012, p. 13). Therefore, these data collectors typically perform *individual-level BDA use* (P2a). The second role is data

Table 3
Supporting Evidence for Propositions.

Category	Proposition	Type of Relation ¹	Source	BDA Specific ²	Non-BDA Specific ²
BDA Value Chain	P1. BDA system capabilities influence BDA use.	CL	System use & BVIT	Ain et al. (2019); Arnott et al. (2017); Aversa et al. (2018); Bowers et al. (2017); Chae (2014); Constantiou and Kallinikos (2015); Dong and Yang (2020); Enholm et al. (2022); Fantini and Narayandas (2023); Fink et al. (2017); Ganju et al. (2022); Ghasemaghaei and Turel (2022); Goyal et al. (2012); Grover et al. (2018); Günther et al. (2017); Joshi et al. (2021); Lebovitz et al. (2022); Lehrer et al. (2018); Mäntymäki et al. (2020); Marabelli et al. (2021); Massa and Testa (2005); Naseer et al. (2021); Popović et al. (2014); Ross et al. (2013); Seddon et al. (2017); Shamim et al. (2019); Shi et al. (2024); Shields (2018); Shollo et al. (2022); Tan et al. (2016); Trieu (2017, 2023); Wang et al. (2019); Wang et al. (2018); Winig (2017); Wu et al. (2020); Xie et al. (2016); Zamani et al. (2021)	Burton-Jones and Grange (2013); Burton-Jones and Volkoff (2017); Kettinger et al. (2013); Lauterbach et al. (2020)
	P2. The characteristics of data collectors, analysers and consumers influence the individual (P2a), collective (P2b) and organisational (P2c) BDA use, respectively.	CL	System use & BVIT	Ain et al. (2019); Arnott et al. (2017); Aversa et al. (2018); Bowers et al. (2017); Brady et al. (2017); Constantiou and Kallinikos (2015); Deng and Chi (2012); Ebrahimi and Matt (2023); Ghasemaghaei and Turel (2022); Günther et al. (2017); Joshi et al. (2021); Lebovitz et al. (2022); Lehrer et al. (2018); Li et al. (2013); Massa and Testa (2005); Namvar et al. (2021); Seddon et al. (2017); Shollo and Galliers (2016); Shollo et al. (2022); Tamm et al. (2020); Tan et al. (2016); Trieu (2017, 2023); Yu et al. (2014)	Burton-Jones and Grange (2013); Burton-Jones and Volkoff (2017); Kang et al. (2012); Lauterbach et al. (2020)
	P3. Individual, collective, and organisational BDA use influences individual (P3a), collective (P3b) and organisational (P3c) goal-oriented activities respectively.	FE	System use & BVIT	Aversa et al. (2018); Bahrami and Shokouhyar (2022); Bowers et al. (2017); Chen et al. (2021); Constantiou and Kallinikos (2015); Goyal et al. (2012); Harris and McDonald (2014); Kowalczyk and Buxmann (2015); Lebovitz et al. (2022); Namvar et al. (2021); Naseer et al. (2021); Nichols (2014); Ross et al. (2013); Seddon et al. (2017); Shields (2018); Shirazi et al. (2022); Someh et al. (2019); Winig (2017); Wu et al. (2020); Zamani et al. (2021)	Alalwan et al. (2014); Burton-Jones and Grange (2013); Burton-Jones and Volkoff (2017); Chen et al. (2020); Kettinger et al. (2013); Sun et al. (2019)
	P4. Individual, collective, and organisational goal-oriented activities influence individual (P4a), collective (P4b) and organisational (P4c) intermediate outcomes respectively.	FE	System use & BVIT	Aversa et al. (2018); Bowers et al. (2017); Ghasemaghaei and Turel (2021); Goyal et al. (2012); Naseer et al. (2021); Rana et al. (2022); Saenz et al. (2022); Seddon et al. (2017); Shamim et al. (2019); Shields (2018); Someh et al. (2019)	Burton-Jones and Grange (2013); Burton-Jones and Volkoff (2017); Chen et al. (2020); Kettinger et al. (2013); Sun et al. (2019)
	P5. Intermediate outcomes influence ultimate outcomes.	CL	System use & BVIT	Ain et al. (2019); Aversa et al. (2018); Grover et al. (2018); Lin et al. (2022); Ramakrishnan et al. (2020); Rana et al. (2022); Saenz et al. (2022); Shields (2018); Someh et al. (2019); Suoniemi et al. (2020); Trieu (2017)	Burton-Jones and Volkoff (2017)

(continued on next page)

Table 3 (continued)

Category	Proposition	Type of Relation ¹	Source	BDA Specific ²	Non-BDA Specific ²
	P6. BDA organisational capabilities influence BDA use.	CL	BVIT	Ain et al. (2019); Aversa et al. (2018); Bowers et al. (2017); Chen et al. (2015); Constantiou and Kallinikos (2015); Enholm et al. (2022); Goyal et al. (2012); Grover et al. (2018); Marabelli et al. (2021); Ransbotham and Kiron (2017); Ross et al. (2013); Shamim et al. (2019); Shields (2018); Shollo et al. (2022); Someh et al. (2019); Trieu (2023); Wang et al. (2019)	Kettinger et al. (2013)
	P7. Complementary organisational resources influence BDA use.	CL	BVIT	Ain et al. (2019); Audzeyeva and Hudson (2016); Bowers et al. (2017); Chen et al. (2015); Constantiou and Kallinikos (2015); Enholm et al. (2022); Goyal et al. (2012); Harris and McDonald (2014); Joshi et al. (2021); Lebovitz et al. (2022); Lehrer et al. (2018); Marabelli et al. (2021); Nichols (2014); Seddon et al. (2017); Shields (2018); Shollo et al. (2022); Someh et al. (2019); Trieu (2017); Wang et al. (2019)	
	P8. Complementary organisational resources influence goal-oriented activities.	CL	BVIT	Aversa et al. (2018); Kowalczyk and Buxmann (2015); Lebovitz et al. (2022); Mäntymäki et al. (2020); Ross et al. (2013); Seddon et al. (2017)	
BDA Learning Loop	P9. Intermediate outcomes trigger BDA learning and influence elements within the BDA value chain (i.e., BDA users and tasks, BDA system capabilities).	FE (1st) & CL (2nd)	System use & BVIT	Bowers et al. (2017); Deng and Chi (2012); Goyal et al. (2012); Grover et al. (2018); Joshi et al. (2021); Li et al. (2013); Massa and Testa (2005); Namvar et al. (2021); Ross et al. (2013); Seddon et al. (2017); Shields (2018); Shollo and Galliers (2016)	Barki et al. (2007); Benlian (2015); Burton-Jones and Grange (2013); Lauterbach et al. (2020)
Context	P10. Internal and external environment moderates the relationships in Proposition 1 to 9.	CL	System use & BVIT	Audzeyeva and Hudson (2016); Aversa et al. (2018); Chae (2014); Chen et al. (2015); Constantiou and Kallinikos (2015); Dong and Yang (2020); Ganju et al. (2022); Grover et al. (2018); Lehrer et al. (2018); Mikalef and Krogstie (2020); Müller et al. (2018); Naseer et al. (2021); Park et al. (2017); Wang et al. (2019)	Kettinger et al. (2013)

* Blank cells denote that there was no paper in our review investigating the propositions.

¹ CL refers to a cross-level relationship. FE refers to a functionally equivalent relationship.

² BDA specific refers to articles where the BDA system was examined in conjunction with system use and/or business value. Non-BDA specific refers to articles that examine both system use and business value of IT other than BDA.

analysers who perform data analysis tasks and collectively use the system to process outputs generated by data collectors along with data automatically collected from various sources to support data consumers (Burton-Jones & Volkoff, 2017). Due to the collective and dependent nature of their data analysis tasks, data analysts generally engage in *collective-level BDA use* (P2b). The third role is data consumers who perform data consumption tasks, typically serving in executive and managerial roles, and are focused on meeting organisational goals (Burton-Jones & Volkoff, 2017; Murphy et al., 2012). Therefore, data consumers typically engage in *organisational-level BDA use* (P2c).

In addition to the roles, user characteristics (e.g., expertise, motivation, and goals) and task characteristics (e.g., task structure and complexity) influence the use of BDA (Deng & Chi, 2012; Lauterbach et al., 2020; Mikalef & Krogstie, 2020; Seddon et al., 2017). In terms of user characteristics, literature on the users of BDA has focused predominately on the characteristics of data analysts and data consumers. Data analysts require a combination of analytical and managerial skills to create analytic products (Mikalef & Krogstie, 2020). While the literature often indicates the need for data consumers to leverage their “previous experience and expertise, common sense and contextual knowledge” (Shollo & Galliers, 2016, p. 353) when using BDA, their personal biases can impede the use of BDA insights (Fantini and Narayandas, 2023). In addition, both data analysts and data consumers need to understand the black box behind how the analytics were generated to minimise biases present in leveraging data-driven insights (Joshi et al., 2021). While the characteristics of data collectors have not commonly been examined in BDA literature, broader IS literature consistently highlights the need

for skilled and motivated users to enhance the use of systems (Benlian, 2015).

In comparison to user characteristics, less attention has been placed on task characteristics. According to Lauterbach et al. (2020), task complexity stems from the extent to which the system supports the accomplishment of tasks. Greater task complexity results in counteracting effects on effective use (Lauterbach et al., 2020) and noncompliant use of BDA (Ghasemaghahi & Turel, 2022). Task structure is also an important task characteristic that aligns with users' roles, and how they use BDA. For instance, operational tasks (commonly performed by data collectors) are more structured (i.e., "repetitive and routine") whereas strategic tasks (commonly performed by data consumers) are more unstructured (i.e., "no fixed method of handling it and... made on a non-repetitive basis") (Işık et al., 2013, p. 16).

In the BDA context, there is dependency of tasks; data consumption tasks depend on the completion of data analysis tasks; data analysis tasks depend on the completion of data collection tasks; and the completion of both these tasks requires the use of BDA. As such, "the actions and interactions among individuals that generate the collective phenomenon" occur through users using BDA (Burton-Jones & Gallivan, 2007, p. 661), indicating there is no structural relationship among each category of BDA users and tasks. Therefore, the relationship between these two constructs (i.e., *BDA user and tasks* and *BDA use*) is cross-level, and we thus propose:

Proposition 2. *The characteristics of data collectors, analysers, and consumers and their tasks influence the individual (P2a), collective (P2b) and organisational (P2c) BDA use, respectively.*

Following *BDA use*, *goal-oriented activities* are performed. Actionable decisions are a critical element to complete a BDA user's task (Seddon et al., 2017). This is consistent with recent research on effective use of BDA which implicitly includes task-related activities within the concept of informed action (Trieu et al., 2022). From a BVIT perspective, *BDA use* is usually implicit when it comes to examining *goal-oriented activities*.

Torres et al. (2018, p. 825) conceptualise BDA capabilities as sensing and seizing capabilities that "identify and shape opportunities through the gathering and analysis of data" and integrate and interpret this analysis to form decisions and allocate resources to support the execution of decisions. This implicitly indicates that goal-oriented decisions are informed by insights generated through *BDA use*. Torres et al. (2018, p. 825) also propose business process change capabilities as an organisation's "ability to effectively alter its business processes to exploit identified opportunities and avoid threats... to translate the action plan developed through BI&A (business intelligence and analytics) seizing capabilities into improved ordinary capabilities". This implies that goal-oriented tasks are performed after the decisions are made.

From a multilevel perspective, an individual can incorporate insights gained from the system to make decisions individually (Burton-Jones & Volkoff, 2017) (P3a). Additionally, if coordination is required, an individual can incorporate insights into team decision-making, which determines appropriate team actions (Burton-Jones & Volkoff, 2017) (P3b). At an organisational level, through using data integrated from different domains to access "critical information in an understandable format", data-driven decisions can be made (Shields, 2018, p. 112), and used to inform or guide intended value-creation actions (Seddon et al., 2017) (P3c). Thus, we propose:

Proposition 3. *Individual, collective, and organisational BDA use influence individual (P3a), collective (P3b) and organisational (P3c) goal-oriented activities respectively.*

Goal-oriented activities are necessary for positive outcomes to be generated (Burton-Jones & Volkoff, 2017; Seddon et al., 2017). From a system use perspective, according to Burton-Jones and Volkoff (2017) and Strong et al. (2014), actualising interrelated and interacting affordances can lead to the attainment of immediate concrete outcomes. An affordance in an organisation is "the potential for goal-oriented behaviour to achieve an immediate concrete outcome" (Strong et al., 2014, p. 70). It can be actualised by taking actions through "using the technology to achieve the outcome in support of organisational goals" (Strong et al., 2014, p. 70). In the BDA context, BDA users need to not only interact with a BDA system and leverage the system outputs but also engage in decision- and task-related activities to achieve intermediate outcomes. As such, BDA users need to perform both *BDA use* and *goal-oriented activities* to achieve *intermediate outcomes*. From a BVIT perspective, organisations which possess BDA-enabled sensing capabilities are able "to sense the environment for decision-making" to make changes and improve their marketing mix (e.g., marketing innovation and methods) (Saenz et al., 2022, p. 2).

Literature has typically examined the relationship between *goal-oriented activities* and *intermediate outcomes* at an individual and organisational level and has seldom explored its relationship across multiple levels (Aversa et al., 2018; Burton-Jones & Volkoff, 2017). At an individual level, if the representations a user obtains from a system "reflect the domain being represented", users are more likely to take actions to achieve their goals, and these actions, in turn, improve their task efficiency and effectiveness (Burton-Jones & Grange, 2013, p. 642) (P4a). At a collective level, goal-oriented actions taken to achieve intermediate outcomes span multiple levels (Burton-Jones & Volkoff, 2017). Aversa et al. (2018) identify that although final decisions can be made by a single decision maker, a group of people is involved in the data analysis process, which suggests a collective level of decision-related activities (P4b). At an organisational level, decisions informed by insights from the use of BDA are not guaranteed to result in positive outcomes, because only when these decisions lead to beneficial actions and these actions are appropriately performed, can these outcomes be created (Seddon et al., 2017). According to Seddon et al. (2017), making informed decisions from data-driven insights and performing resultant actions ultimately result in positive organisational outcomes (P4c). Thus, we propose:

Proposition 4. *Individual, collective, and organisational goal-oriented activities influence individual (P4a), collective (P4b) and organisational (P4c) intermediate outcomes respectively.*

According to Strong et al. (2014), *individual-level intermediate outcomes* aggregate to form *organisational-level intermediate outcomes*,

which eventually contribute to attaining *organisational ultimate outcomes*, termed ‘organisational goals’ in their research. This relationship is elaborated by Burton-Jones and Volkoff (2017) who indicate that the *ultimate outcomes* of effective use are achieved through a connected set of *intermediate outcomes*, which are generated from users across an organisation. For example, “identifying new business opportunities and capitalizing on them” (i.e., *organisational intermediate outcomes*) is likely to improve an organisation’s economic value (i.e., *ultimate outcomes*) (Ramakrishnan et al., 2020, p. 31). Likewise, different types of *intermediate outcomes* (e.g., decision- and task-related outcomes) contribute to different types of *ultimate outcomes* (Someh et al., 2019). This is supported by Grover et al. (2018) who highlight that different value targets of a BDA system lead to different organisational impacts. For instance, Aversa et al. (2018) identify that poor decision performance negatively influences an organisation’s economic and strategic value. Saenz et al. (2022) identify that improvements in *intermediate outcomes*, such as marketing innovation, can lead to improvements in *ultimate outcomes*, such as market performance. However, the realisation of *ultimate outcomes* may be delayed because *intermediate outcomes* take time to accumulate (Trieu, 2017).

The relationship between *intermediate outcomes* and *ultimate outcomes* is cross-level. This has been shown by past research defining *intermediate outcomes* at multiple levels, and *ultimate outcomes* at the organisational level (Burton-Jones & Volkoff, 2017; Strong et al., 2014). Thus, we propose:

Proposition 5. *Intermediate outcomes influence ultimate outcomes.*

As argued by Grover et al. (2018), in order to realise BDA system’s strategic business value (i.e., *ultimate outcomes*), an organisation needs to build *BDA organisational capabilities*. Different from *BDA system capabilities*, organisational capabilities are the BDA-specific abilities to effectively implement and adopt BDA systems for the generation of ultimate outcomes. To build capabilities, an organisation must establish a BDA infrastructure, including big data assets, analytics portfolios, and human talent, as well as develop the capabilities to integrate, manage, share and analyse big data (Chen et al., 2015; Grover et al., 2018). This requires the organisation to manage how big data is “created, stored, processed and accessed within and throughout organisational boundaries” to address potential issues (Mikalef et al., 2020a, p. 3), such as “security, privacy, regulation compliance, or liability of big data” (Grover et al., 2018, p. 399). To leverage the analytics portfolios, the organisation needs to decide on the “data aggregation, transformation, distributed computing, tool and analytics models” to be employed (Grover et al., 2018, p. 399). BDA professionals also need to be present throughout the lifecycle from designing a BDA strategy to interpreting BDA outputs. Subsequently, the developed capabilities can be realised and contribute to different intermediate outcomes through different value creation mechanisms, which implicitly describe *BDA use*. Lastly, data-driven culture should be promoted as it can contribute to identifying and producing potential BDA value (Grover et al., 2018). Although “adopting evidence-based decision making is a difficult cultural shift... once companies have made the cultural change, they usually do not go back, and their operating improvements are not easily replicated by competitors” (Ross et al., 2013, p. 4).

Regarding the levels of *BDA organisational capabilities* and *BDA use*, the former operates at an organisational level while the latter operates at multiple levels. As such, the relationship between these constructs is theorised as cross-level, indicating that *BDA organisational capabilities* impacts a single user, a group of interdependent and goal-oriented users, or an organisation using a BDA system. Thus, we propose:

Proposition 6. *BDA organisational capabilities influence BDA use.*

BDA systems need to be used with *complementary organisational resources* (e.g., people, processes, and technologies) to create value (Mikalef & Krogstie, 2020; Seddon et al., 2017). These resources are not tailored to BDA systems but are related to attaining users’ goals (Xu et al., 2016). Organisations need to ensure there is alignment between organisational strategies and BDA use processes, facilitating users to attain their goals and achieve the desired outcomes (Joshi et al., 2021). BDA systems also need to be complemented with other technological resources (e.g., customer relationship management systems) in order to synergistically create positive outcomes to attain organisational goals (Someh et al., 2019). Furthermore, non-technological resources (e.g., human and financial) play an important role. When non-technological resources are lacking, the effective use of BDA systems may be hindered due to increasing workloads (Lebovitz et al., 2022). In addition to the aforementioned resources, organisational knowledge including domain knowledge can facilitate the use of BDA (Wang et al., 2019). According to Trieu (2017), *complementary organisational resources* in the form of non-BI investments serve to complement BI investments and influence the BI conversion and, in turn, BI use processes. Although the BI conversion process is not examined in our research, we consider the effects of *complementary organisational resources* on *BDA use*.

From a multilevel perspective, *complementary organisational resources* operate at an organisational level, while *BDA use* operates at multiple levels. As such, the relationship is cross-level, indicating the organisational-level resources have effects on the use of BDA by a single user, a group of interdependent and goal-oriented users, or an organisation. Thus, we propose:

Proposition 7. *Complementary organisational resources influence BDA use.*

The literature also recognises the importance of *complementary organisational resources* for the execution of *goal-oriented activities*. For example, “business rules are the mechanism for specifying what actions should be taken in a given circumstance” (Ross et al., 2013, p. 7). As such, it is important to ensure that goal-oriented activities are aligned “with the strategic objectives of the company” (Ross et al., 2013, p. 7). In addition, Seddon et al. (2017) demonstrate that different resources contribute to different BDA value-creating entities where analytical resources (i.e., *BDA system capabilities* and *BDA organisational capabilities*) enable *BDA use*, and organisational resources (i.e., *complementary organisational capabilities*) enable decision making and other actions (i.e., *goal-oriented activities*). According to Seddon et al. (2017), an organisation can intend to perform actions using organisational resources to create organisational benefits. The definitions of resources and actions in Seddon et al. (2017) align with how we define *complementary*

organisational resources and goal-oriented activities respectively in our framework.

Consistent with the multilevel theorising of *complementary organisational resources* and *BDA use*, the relationship between these constructs is cross-level. This indicates that the organisational-level resources have effects on task- and decision-related activities performed by a single user, a group of interdependent and goal-oriented users, or an organisation. Thus, we propose:

Proposition 8. *Complementary organisational resources influence goal-oriented activities.*

BDA learning loop

Previous studies have demonstrated that learning is one approach to iteratively improve system use (Barki et al., 2007; Burton-Jones & Grange, 2013; Deng & Chi, 2012; Lauterbach et al., 2020; Strong et al., 2014). Most studies in our review focused on different aspects of learning and provide limited discussion of the learning process. By synthesising these studies, we propose that the BDA learning loop can positively affect *BDA use* through influencing *BDA users and tasks* and *BDA system capabilities* in an ongoing manner and eventually contribute to generating *ultimate outcomes* (Fig. 4). The loop begins when a user: 1) gains feedback from *intermediate outcomes*, which 2) triggers *BDA learning* across different levels targeted at improving *BDA users and tasks* and *BDA system capabilities*. This, in turn, 3) influences *BDA use*, which as theorised in the BDA value chain contributes to *ultimate outcomes*. Below, we focus on the first two processes, as the third has been discussed in the value chain propositions.

According to Burton-Jones and Grange (2013) and Strong et al. (2014), learning is initiated through users gaining feedback from their intermediate outcomes. In the context of BDA use, these *intermediate outcomes* reflect users' task and decision-making performance resulting from their *goal-oriented activities* that stem from *BDA use*. *BDA learning*, as a multilevel construct, can be initiated at any level (Wiewiora et al., 2019). *Individual BDA learning* can be triggered from an individual self-identifying, through reflection-in-action, the need to improve how they perform their tasks (Burton-Jones & Volkoff, 2017). The knowledge an individual acquires through learning can be communicated and shared to groups over time, and institutionalised at an organisational level (Shollo & Galliers, 2016). In addition to bottom-up learning, *organisational BDA learning* can be triggered when the organisation identifies systematic deficits in their BDA knowledge as reflective in commonalities in poor decision or task outcomes. Organisational activities (e.g., training) can then be performed to deliver knowledge to groups and individuals (Jha et al., 2020).

Through engaging in *BDA learning*, improvements in *BDA users and tasks* and *BDA system capabilities* can result, facilitating *BDA use*. In terms of improvements to *BDA users and tasks*, Barki et al. (2007, p. 176) identify that individuals, when learning how to use systems, modify their use behaviours "via increased system knowledge and mastery". This can result from individuals communicating with others to exchange information and independently exploring knowledge (Barki et al., 2007). Additionally, Burton-Jones and Grange (2013, p. 644) describe that users can learn "the system, the domain it represents, the extent to which it faithfully represents the domain, or how to engage in more informed actions" to improve system use. This is substantiated by Deng and Chi (2012) and Benlian (2015) who express the need to improve domain knowledge in addition to knowledge of the BDA system and data. In terms of improvements to *BDA system capabilities*, Barki et al. (2007) also identify that learning can be targeted at adapting systems to better fit the

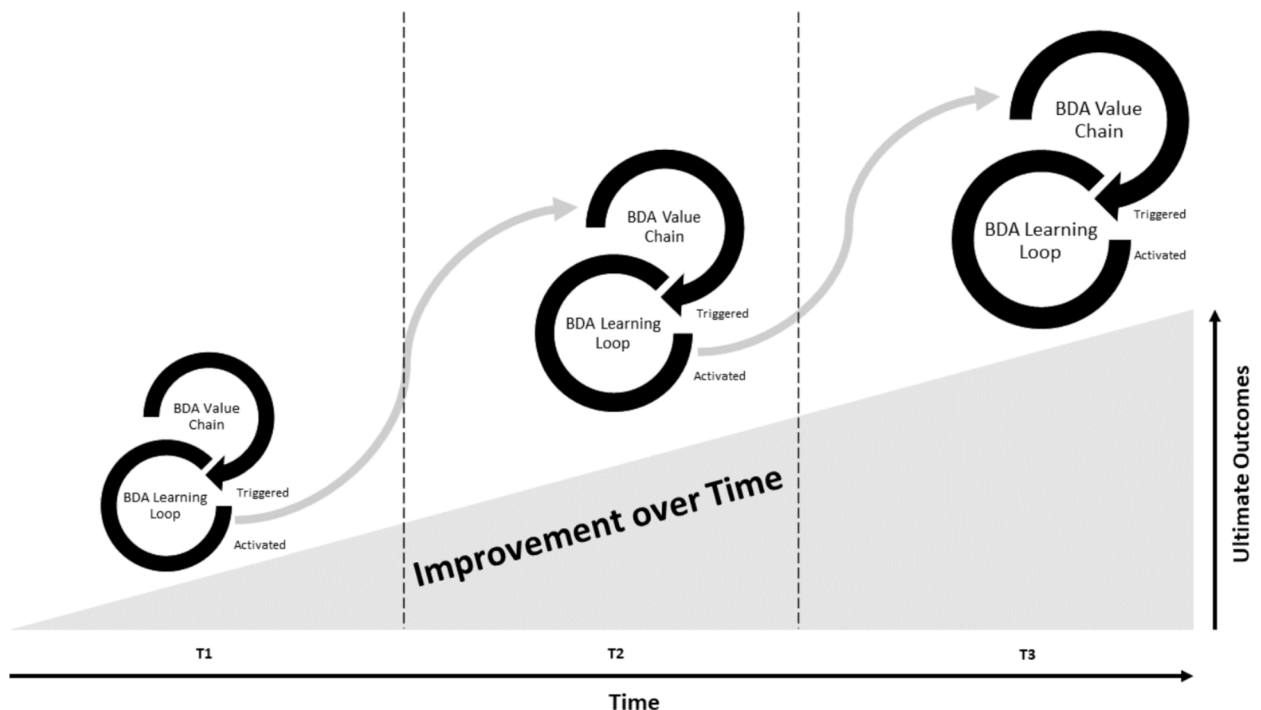


Fig. 4. BDA Learning Loop.

needs of the individuals, the processes, and the organisation. Seddon et al. (2017) further identify that organisational learning can improve the quality of data within the system.

From a multilevel perspective, both *BDA system capabilities* and *BDA users and tasks* operate at a single level, and *BDA learning* operates at multiple levels. Thus, the relationship of *BDA learning* and *BDA system capabilities*, and the relationship of *BDA learning* and *BDA users and tasks* are both cross-level. Thus, we propose:

Proposition 9. *Intermediate outcomes trigger BDA learning and influence elements within the BDA value chain (i.e., BDA users and tasks, BDA system capabilities).*

Contextual factor moderating the relationships

Emerging insights suggest that internal factors (e.g., organisational inertia, organisational size) and external contextual factors (e.g., industry and country characteristics, competition, government regulation) moderate the relationships within the BDA value chain. For example, Grover et al. (2018) highlight the need to consider the *internal and external environment* as a moderator that catalyses the conversion of BDA system to BDA value (i.e., *ultimate outcomes*). Similarly, Dong and Yang (2020) identify that organisational size moderates the relationship between *BDA system capabilities* and organisational performance. Chen et al. (2015) further evidence that the relationships between the use of BDA and productivity and business growth are moderated by the dynamism of the external environment. The relationship between system capabilities and system use is also moderated by external factors, with Ganju et al. (2022) evidencing that external government programs moderate the relationship between BDA system capabilities and effective documentation practices. These examples provide initial insights to support the moderating effect of *the internal and external environment* on the relationships present within the BDA value chain. However, we did not find examples in the literature to support the moderating effects on all relationships present. We posit that this is likely due to literature focusing broadly on the whole value chain as opposed to unpacking the BDA value chain into its constituent elements. Consistent with Müller et al. (2018) and Grover et al. (2018), we posit that it is likely that contextual factors will moderate the relationships within the framework.

Proposition 10. *Internal and external environment moderates the relationships in Proposition 1 to 9.*

Discussion

System use is critical to understanding how BDA systems create value. Although the business value of IT (BVIT) and system use have been well studied in isolation, studies integrating them are relatively scarce. Through synthesising these literature streams, we developed the BDA-UV framework. This framework places the BDA value chain within the BDA contextual environment and describes how *BDA system capabilities* – through the use of BDA systems, execution of *goal-oriented activities*, and the resultant accumulation of *intermediate outcomes* – result in *ultimate outcomes*. Consistent with the system use literature, the BDA-UV framework further identifies that *BDA use* is influenced by *BDA system capabilities*, *BDA users and tasks*, and the effects of *BDA learning*. Drawing on BVIT literature, the BDA-UV framework also recognises the role of *BDA organisational capabilities* and *complementary organisational resources* in facilitating *BDA use*. The BDA-UV framework also provides initial insights supporting the moderating role of the contextual *internal and external environment*.

In addition to being examined separately, BVIT literature and system use literature have focused on different levels of analysis, with BVIT examining the organisational level and system use examining the individual level, leading to calls for a multilevel perspective (Burton-Jones & Gallivan, 2007; Zhang et al., 2019). By rigorously considering multilevel theorising guidelines (Burton-Jones & Gallivan, 2007; Zhang et al., 2019), we developed the BDA-UV framework which identifies single-level and multilevel constructs that form cross-level and functionally equivalent relationships.

Below, we discuss the BDA-UV framework from BVIT, system use, and BDA perspectives by comparing constructs and seminal models sharing a similar intent, followed by future research opportunities.

BVIT perspective

Below, we compare the BDA-UV framework with the seminal, integrative BVIT model of Melville et al. (2004) and BDA models building on it (e.g., Grover et al., 2018; Trieu, 2017). First, aligned with an RBV perspective of BVIT (e.g., Melville et al., 2004), the BDA-UV framework demonstrates that the more BDA systems are valuable, rare, costly to imitate, and organisationally embedded, the more likely they can create value for the organisation. Second, *BDA use* and *goal-oriented activities* in the BDA-UV framework address how BDA systems are applied within business processes, extending the RBV perspective (Melville et al., 2004). This can impact business process performance as an *intermediate outcome* and organisational performance as an *ultimate outcome*, which is consistent with Melville et al. (2004), Trieu (2017), and Grover et al. (2018). Third, we highlight two ways to complement *BDA system capabilities* to enhance *BDA use*: via *BDA organisational capabilities* and via *complementary organisational resources*. This aligns with ideas on synergistic relationships between different types of resources and capabilities for creating business value with IT (e.g., Nevo & Wade, 2010) and data analytics (e.g., Dong & Yang, 2020; Someh et al., 2019).

Finally, by including the *BDA learning loop*, we extend the RBV perspective on the BVIT (e.g., Chae et al., 2014; Melville et al., 2004) with a DCT perspective. Dynamic capabilities can positively affect operational IT and organisational capabilities by altering the firms' resource base (Steininger et al., 2022). The evolution and improvement of capabilities is guided by organisational learning (Wheeler, 2002). While others have also extended RBV with DCT for the business value of BDA (e.g., Corte-Real et al., 2017), the BDA-UV framework explicitly specifies the role of multilevel BDA learning, drawing attention to learning as a dynamic process (Crossan

et al., 1999) and acknowledging different learning paths and interactions (Wiewiora et al., 2019).

Researchers interested in the business value of BDA should examine the actions of users to achieve their goals (*BDA users and tasks*, *BDA use*, and *goal-oriented activities*), the short- and long-term value generated from their actions (*intermediate* and *ultimate outcomes*), and the BDA and organisational factors influencing value (*BDA system capabilities*, *BDA organisational capabilities* and *complementary organisational resources*). First, by including the characteristics and actions of users that are necessary to generate outcomes, the BDA-UV framework allows researchers to investigate how the use of BDA and subsequent activities generate positive decision- and task-related outcomes. This links to research on microfoundations, which addresses the importance of individuals and their interactions for organisational performance (Felin et al., 2012). Future research should, therefore, seek to understand how the microfoundations of users' goal-oriented behaviours – emanating from their characteristics and use of BDA – impact organisational outcomes. Second, researchers should seek to understand the multilevel dynamics surrounding how individual, collective, and organisational levels of *intermediate outcomes* accumulate over time to generate organisational-level *ultimate outcomes*. It is widely acknowledged that lag effects can delay the impact of IT investments (e.g., Schryen, 2013; Steininger et al., 2022). Future research can specifically address the temporal role of bottom-up and top-down effects across *intermediate*, *decision- and task-related outcomes* and what this means for the *ultimate outcomes*. Third, researchers need to further understand the BDA and organisational factors influencing BDA business value: *BDA system capabilities*, *BDA organisational capabilities*, and *complementary organisational resources*. This will involve examining both the synergetic relationships (e.g., Dong & Yang, 2020; Nevo & Wade, 2010; Someh et al., 2019) and tensions between resources and capabilities for creating business value (Günther et al., 2017).

System use perspective

Consistent with Burton-Jones and Straub's (2006) seminal conceptualisation of system use, the BDA-UV framework identifies the system (*BDA system capabilities*), and user and task characteristics (*BDA users and tasks*) that influence *BDA use*, which in turn influences *intermediate outcomes*. When users use BDA systems and perform *goal-oriented activities* to complete their tasks, task-related outcomes can occur, which can trigger *BDA learning* to improve *BDA use*. In addition to extending system use literature by incorporating organisational factors from BVIT (i.e., *BDA organisational capabilities*, *complementary organisational resources*), we further contribute to system use literature in two ways. First, we reposition the role of data analysts as a form of users who play a central and ongoing role in using BDA to create analytical products. This differs to extant literature that views data analysts as part of the technical team who are isolated and external to the system use process (Burton-Jones & Volkoff, 2017; Trieu et al., 2022). Second, the BDA-UV framework extends beyond the largely single-level perspective of system use often apparent in literature to recognise the multilevel system use process and the top-down and bottom-up dependencies.

Researchers interested in the use of BDA should investigate, in a multilevel fashion, the antecedents of *BDA use* (*BDA system capabilities*, and *BDA users and tasks*), different use behaviours (*BDA use*), and subsequent actions taken (*goal-oriented activities*) to generate value (*intermediate outcomes*), as well as learning effects (*BDA learning*) to facilitate *BDA use*. Considering the use of BDA is central to generating value from BDA, future research needs to robustly conceptualise *BDA use*. In this review, we identified many types of system use (e.g., routine use, innovative use, adaptive use, effective use), with effective use increasing in prominence due to its focus on informed actions (Trieu et al., 2022). Future research needs to understand how these system use behaviours are related as well as their outcomes. Research also needs to consider whether new types of system use are required given the informing potential of BDA (Zuboff, 1988) and the increased delegation of decision-making power to the system rather than to humans (Grønsund & Aanestad, 2020; Mayer et al., 2020). For industries that heavily involve humans in the data collection, analysis, and consumption activities, future research also needs to be conducted into learning how to improve the use of BDA to “enhance their regular practices”, “support adaptation and innovation in an ever-changing world”, and “optimise their use of existing resources” (Sheikh et al., 2021, p. xv). Improving *BDA use* and *learning* requires targeted longitudinal and multilevel studies to understand how *BDA use* at one level influences other use behaviours at other levels, which has a cascading effect on *goal-oriented activities*, *intermediate outcomes*, and *BDA learning*.

BDA perspective

Below, we discuss the BDA-specific elements within the BDA-UV framework and compare this framework with seminal models that have examined the use and business value of BDA. In addition, we provide avenues for future research to engage in contextual and multilevel theorising to further unpack the role of use for the business value of BDA.

Overall, the BDA-UV framework contains elements – such as big data and information quality which are dimensions of *BDA system capabilities* (Grover et al., 2018) – that are specific to BDA systems and their path to value creation. As big data is collected from various sources, it requires pre-processing to ensure its quality for subsequent analysis (Grover et al., 2018), which influences the effective use of information (Kettinger et al., 2013). The users and tasks within the BDA-UV framework further reflect the BDA context, recognising the different types of roles and tasks of BDA users from collection to analysis to consumption, which provides a more complete view of the use of BDA. *BDA-specific organisational capabilities* are also identified, including: data governance (Mikalef et al., 2020a), establishment and development of BDA organisational infrastructure (Chen et al., 2015; Grover et al., 2018), and data-driven culture (Grover et al., 2018). When it comes to *intermediate outcomes* generated from *BDA use* and *goal-oriented activities*, decision-related activities and outcomes are highly relevant to the BDA context as the system is designed to discover insights to support and improve decision making (Chen et al., 2012; Krishnamoorthi & Mathew, 2018; Lavallo et al., 2011).

In addition to the BDA-specific elements described above, we compare our framework with two notable models – Seddon et al.

(2017) and Trieu (2017) – that focus on use and business value within the BDA context. Seddon et al. (2017) developed a model from the senior management perspective that explicates three pathways for how short-term (e.g., based on the implementation of BDA projects) and long-term factors (e.g., based on the use of analytics tools) contribute to organisational benefits. Compared with Seddon et al. (2017), our framework integrates a multilevel perspective of *BDA use*. This enables more in-depth insights of the mechanisms behind the pathways through identifying single-level (e.g., *BDA organisational capabilities*) and multilevel factors (e.g., *BDA learning*) that influence *BDA use*. Trieu (2017), in following Soh and Markus (1995), identified three data analytics value creation processes, which explain how data analytics investments can result in data analytics assets (the conversion process), which can then be turned into data analytics impacts (the use process), and can ultimately contribute to organisational performance (the competitive process). In addition, Trieu (2017) also identifies the influence of contextual factors and latency effects. Compared with Trieu (2017), the BDA-UV framework delves more deeply into the use process, demonstrating its complex, multi-actor and multilevel nature and its impacts on subsequent actions and outcomes. Additionally, the BDA-UV framework also recognises the prominence of the *BDA learning loop* in ongoing efforts to improve *BDA use*. The efforts to improve *BDA use* are also complemented by *BDA organisational capabilities* and *complementary organisational resources*. These novel insights serve to illuminate specific factors that can be improved and mechanisms that can be implemented to realise the value of BDA.

When investigating the use and value of BDA, future research should examine the BDA-UV framework in conjunction with contextual theorising (Hong et al., 2014) to further unpack the context-specific antecedents of *BDA use*. This will involve identifying what is unique about the individual and organisational resources and capabilities necessary for BDA. In our review, Trieu et al. (2022) is one of the few studies that sought to robustly contextualise data analytics specific system capabilities (e.g., data analysis, queries, reports) and organisational capabilities (e.g., data dictionary, business rules, data-drive culture). When engaging in contextual theorising, future research should also seek to identify how the instantiations of the constructs within the BDA-UV framework differ between BDA and traditional IT. Within the BDA context, understanding how analytical insights can shape data-driven decisions is of central importance because, as Watson (2014, p. 1252) notes, value from BDA is “created only when the data is analysed and acted upon”. The theory of effective use provides an initial understanding by hypothesising that informed actions are shaped by transparent interaction and the quality of representations (Burton-Jones & Grange, 2013) but provides little guidance into how specific visualisations of analytics shape decision. This will become increasingly important as the sophistication of analytics techniques increase.

In addition, the BDA-UV framework in providing a robust multilevel perspective of the role of use for the business value of BDA, also offers avenues for future research to extend beyond the dominant single-level perspective apparent in system use and BVT literature. For example, to examine the phenomenon in a single setting, researchers can explore bottom-up effects (e.g., individual level to collective level and collective level to organisational level) on multilevel relationships between and within multilevel constructs in the BDA context. This allows researchers to answer the questions of whether and how *BDA intermediate outcomes* flow from an individual level to an organisational level and accumulate to *ultimate outcomes*. For multiple settings, researchers can explore a top-

Table 4
Future Research Agenda.

Perspective	Future Research Directions	Possible Research Questions
BVT perspective	1. The actions taken by users to achieve their goals (<i>BDA users and tasks, BDA use, goal-oriented activities</i>) 2. The short- and long-term value generated from taking the actions (<i>intermediate outcomes and ultimate outcomes</i>) 3. The BDA and organisational factors influencing BDA business value (<i>BDA system capabilities, BDA organisational capabilities and complementary organisational resources</i>)	1. How can the microfoundations of users' goal-oriented behaviours emanating from their characteristics and use of BDA impact the attainment of organisational outcomes? 2. How can accumulation of short-term outcomes from BDA impact the attainment of the long-term organisational outcomes? 3. How can potential synergies and tensions between BDA system capabilities, BDA organisational capabilities, and complementary organisational resources impact the attainment of organisational outcomes?
System use perspective	1. The elements of system use (<i>BDA system capabilities, and BDA users and tasks</i>) 2. The types of <i>BDA use</i> behaviours and subsequent actions (<i>goal-oriented activities</i>) conducive to value realisation 3. The types of learning (<i>BDA learning</i>) that facilitates <i>BDA use</i> behaviours and value.	1. How are the roles of users and tasks, as elements of system use, evolving in the presence of BDA? 2. How do different types of BDA use shape the activities of themselves and other users to achieve their goals? 3. What forms of learning enhance BDA use behaviours and enhance value creation?
BDA perspective	1. The dimensions of <i>BDA system capabilities, BDA users and tasks, BDA organisational capabilities, and complementary organisational resources</i> that influence <i>BDA use</i> 2. The differences between BDA and traditional IT across all constructs within the BDA-UV framework. 3. The ways BDA insights are derived and presented, and their influence on informed decisions (<i>goal-oriented activities</i>)	1. What is unique about individual and organisational resources and capabilities necessary for BDA? 2. What are the implications of functionally-oriented BDA compared to organisationally-oriented IT for business value? 3. How does the design of BDA influence informed decision-making and value creation?
Multilevel perspective	1. The bottom-up effects (e.g., individual level to collective level and collective level to organisational level) on multilevel relationships between and within multilevel constructs in the BDA context 2. The top-down effects (e.g., organisational level to collective level and collective level to individual level) on multilevel relationships between and within multilevel constructs in the BDA context 3. The longitudinal unfolding of bottom-up and top-down multilevel relationships in the BDA context.	1. What are the different patterns of individual actions when interacting with BDA and how do they influence collective and organisational level outcomes? 2. How do organisational actions enable and constrain individual users to achieve their outcomes from BDA? 3. How does top-down and bottom-up learning directed towards BDA unfold over time and impact user behaviour towards favourable outcomes?

down effect (e.g., organisational level to collective level and collective level to individual level), such as identifying what contextual or organisational-level factors facilitate or hinder users' effective use of BDA. In addition to different types of effects, researchers can also explore different types of multilevel relationships between constructs, in terms of cross-level and functionally equivalent relationships.

Summary of future research directions

Informed by the insights generated from our systematic review and elaborated upon in the preceding sections of the Discussion, we propose a research agenda (Table 4). The research agenda provides high-level future research directions accompanied by potential future research questions (to aid their operationalisation) of relevance to BVIT, system use, BDA, and multilevel perspectives. From a BVIT perspective, future research can both extend and deepen BVIT by attending to the microfoundations of BDA value creation, examining the longitudinal unfolding of ultimate outcomes from intermediate outcomes, and unpacking the synergies and tensions amongst organisational, technical, and user factors. From a system use perspective, future research can deepen the understanding of the components of system use (i.e., system, user, task) by rethinking the traditional roles of users, and integrate system use literature by examining how different forms of system use influence each other. From a BDA perspective, future research can further contextualise BDA by teasing apart the differences between BDA and other IT resources and capabilities as well as the influence of BDA on decision-making. From a multilevel perspective, the BDA-UV framework provides an opportunity to explore the top-down and bottom-up relationships often hidden in single-level research. Overall, as Table 4 demonstrates, substantial research is needed to better understand the role of system use for the business value of BDA.

Conclusion

This study is motivated by the need to understand the interplay between BDA, system use, and BVIT. It makes several theoretical and practical contributions. The first major theoretical contribution is the development of the multilevel BDA-UV framework that integrates theoretically relevant constructs from both system use and BVIT studies, building on insights from multiple disciplines, which are suitable for the BDA context. Second, the BDA-UV framework demonstrates the critical role of *use* for *BDA systems* to generate *ultimate outcomes* and distinguishes between two types of outcomes (i.e., *intermediate outcomes* and *ultimate outcomes*) that have seldom been recognised in prior work. This provides an opportunity for future research to examine the intertwined relationships between *BDA system capabilities* and *BDA use*, and how *BDA use* ultimately influences outcomes. It also provides the basis for further examination of the dependencies between the two types of outcomes and how they are shaped by *BDA use*. Third, the BDA-UV framework highlights the importance of the BDA learning loop to the BDA value chain through improving the capabilities of users and the BDA system and ultimately improving *BDA use*. Fourth, through including the contextual factors, the BDA-UV framework recognises the potential implications of *the internal and external environment*. Future research can conduct a multiple-case study design in different organisations or industries within the same country or in the same industry but in different countries to gain insights regarding how the environment moderates the proposed relationships. In summary, through integrating the disparate yet commensurate research streams of BVIT and system use as well as leveraging insights from multiple disciplines, we have identified future research directions to encourage research at the intersection of these streams.

For practitioners, the BDA-UV framework offers a comprehensive guide on how organisations can maximise the value derived from their BDA investments. In short, the value of BDA investments is dependent upon how these systems and the insights they produce are utilised to inform decisions and subsequent actions. The use of BDA throughout an organisation is highly interdependent and consequential as the use of BDA by operational staff shapes the quality of analytical insights that are leveraged by others at all levels of the organisation. For the use of BDA to be conducive for value creation, the use must be supported by the necessary system, user, and BDA-specific organisational capabilities as well as complementary organisational resources. Improvements in the use of BDA heavily rely on continual learning activities aimed at advancing both user and BDA system capabilities. In addition, attaining value from BDA further requires a deep contextual understanding of the environmental enablers and constraints that influence how the BDA value chain and BDA learning unfolds.

Specifically, the BDA-UV framework can inform executive managers responsible for the BDA strategy (Grover et al., 2018) by directing their attention to an integrated perspective, which covers the supply-side of developing BDA capabilities as well as the demand-side of the use of BDA and outcomes. For the supply side, it can assist executive management to determine what resources and capabilities should be invested in to better attain the value derived from *BDA use* by considering the organisational factors (i.e., *BDA organisational capabilities* and *complementary organisational resources*) identified in the framework. For the demand side, the BDA-UV framework can assist executive management of organisations adopting BDA to tailor their BDA strategies, which can facilitate *BDA use* and in turn contribute to BDA value creation. Moreover, middle managers can gain valuable insights into how individual and task differences affect the use of BDA and user performance, and identify what actions are required to improve their performance by considering the *BDA learning* effects. In addition, BDA systems can enable the formulation and implementation of strategy by executive management and other stakeholders (Morton et al., 2022). As such, the BDA-UV framework can also provide insights for strategising for executive managers and other stakeholders as the users of BDA.

Despite these contributions, the study has two limitations. First, this is a literature review but, since the research topic is an emerging area, there are a relatively limited number of existing studies incorporating these concepts. However, as this is a developmental review, there is sufficient literature coverage since we have covered "studies that are central or pivotal to a topic area and include a sample that considers all important aspects" (Templier & Paré, 2015, p. 6). Although this literature base provides a robust foundation to conceptualise the BDA-UV framework, it also constrains the conceptualisation to existing theoretical perspectives. For

example, while we did not limit the review to a specific theoretical perspective of BVIT, many of the studies included in the review built upon RBV, DCT, or variants based on these two perspectives. Future research should examine how other theoretical lenses could be used to examine BVIT and its relationship with system use. Second, since the BDA-UV framework is developed from literature stemming from different research streams, questions could be raised about the appropriateness of combining these different theoretical insights. However, although the BVIT and system use literature examine different levels of analysis, they are commensurate because they both assert that technology artefacts in and of themselves are not sufficient for benefit attainment and that other factors need to be considered (the organisational level from a BVIT perspective, and behavioural from a system use perspective). Overall, there are ample opportunities for future research emanating from this work.

CRedit authorship contribution statement

Yi-Ting Yeh: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Rebekah Eden:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Erwin Fieft:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Rehan Syed:** Writing – review & editing, Visualization, Methodology, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix 1. Included Studies: BDA-specificity and Research Stream

Paper	BDA (Business Intelligence/ Business Analytics / Big Data Analytics)	System Use	BVIT (Intermediate Outcomes / Ultimate Outcomes)
Adjerid et al. (2023)	V		V
Adya and Phillips-Wren (2020)	V	V	V
Ain et al. (2019)	V	V	V
Alalwan et al. (2014)		V	V
Arnott et al. (2017)	V	V	
Audzeyeva and Hudson (2016)	V		V
Aversa et al. (2018)	V	V	V
Bahrami and Shokouhyar (2022)	V	V	V
Bao and Bardhan (2022)	V	v	V
Barki et al. (2007)		V	V
Bayraktar et al. (2023)	V		V
Beaudry et al. (2020)		V	V
Benlian (2015)		V	V
Bowers et al. (2017)	V	v	V
Božić and Dimovski (2019)	V	V	V
Brady et al. (2017)	V	V	
Burton-Jones and Grange (2013)		V	V
Burton-Jones and Volkoff (2017)		V	V
Chae (2014)	V	V	
Chatterjee et al. (2020)		V	V
Chatterjee et al. (2021)	V		V
Chatterjee et al. (2023)	V		V
Chen et al. (2012)	V		V
Chen et al. (2015)	V	V	V
Chen et al. (2020)		V	V
Chen et al. (2021)	V	V	
Chen et al. (2022)	V		V
Constantiou and Kallinikos (2015)	V	v	V
Côrte-Real et al. (2020)	V		V
Deng and Chi (2012)	V	V	
Dong and Yang (2020)	V	v	V
Drydakis (2022)	V	V	V
Dwivedi et al. (2023)		V	V

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Paper	BDA (Business Intelligence/ Business Analytics / Big Data Analytics)	System Use	BVIT (Intermediate Outcomes / Ultimate Outcomes)
Ebrahimi and Matt (2023)	V	V	
Elia et al. (2022)	V		V
Enholm et al. (2022)	V	V	V
Fantini and Narayandas (2023)	V	v	V
Fink et al. (2017)	V	V	V
Ganju et al. (2022)	V	v	V
Ghasemaghaei et al. (2017)	V	V	V
Ghasemaghaei et al. (2018)	V		V
Ghasemaghaei (2019b)	V	V	V
Ghasemaghaei and Calic (2019)	V	V	V
Ghasemaghaei and Turel (2021)	V		V
Ghasemaghaei and Turel (2022)	V	V	
Goyal et al. (2012)	V	v	V
Grover et al. (2018)	V	v	V
Günther et al. (2017)	V	v	V
Gupta and George (2016)	V		V
Harris and McDonald (2014)	V	v	V
Horita et al. (2023)	V	v	V
Huang et al. (2020)	V		V
Işik et al. (2013)	V	v	V
Jha et al. (2020)	V	V	V
Joshi et al. (2021)	V	v	V
Kang et al. (2012)		V	V
Kettinger et al. (2013)		V	V
Kohli and Melville (2019)		V	V
Kowalczyk and Buxmann (2015)	V	V	V
Kuegler et al. (2015)		V	V
Lakshmi and Corbett (2023)	V	v	V
Lauterbach et al. (2020)		V	V
Lebovitz et al. (2022)	V	V	
Lehrer et al. (2018)	V		V
Li et al. (2013)	V	V	
Li et al. (2019)		V	V
Lin et al. (2022)	V	V	V
Mandal (2019)	V		V
Mäntymäki et al. (2020)	V	V	
Marabelli et al. (2021)	V	V	
Massa and Testa (2005)	V	v	V
Mikalef et al. (2020a)	V		V
Mikalef and Krogstie (2020)	V		V
Mikalef et al. (2020b)	V		V
Müller et al. (2018)	V		V
Namvar et al. (2021)	V	V	
Naseer et al. (2021)	V	V	V
Nichols (2014)	V	v	V
O'Connor and O'Reilly (2018)		V	V
Oesterreich et al. (2022)	V		V
Otondo (2019)	V	V	V
Pang and Pavlou (2019)		V	V
Papagiannidis et al. (2023)	V	v	V
Park et al. (2017)	V		V
Popović et al. (2014)	V	V	
Popović et al. (2018)	V		V
Raguseo et al. (2021)	V		V
Ramakrishnan et al. (2020)	V	v	V
Rana et al. (2022)	V		V
Ransbotham and Kiron (2017)	V	v	V
Ross et al. (2013)	V	v	V
Saenz et al. (2022)	V		V
Saldanha et al. (2017)	V	V	V
Seddon et al. (2017)	V		V
Shamim et al. (2019)	V	v	V
Shao et al. (2023)	V		V
Shi et al. (2022)	V	v	V

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Paper	BDA (Business Intelligence/ Business Analytics / Big Data Analytics)	System Use	BVIT (Intermediate Outcomes / Ultimate Outcomes)
Shi et al. (2024)	V	V	V
Shields (2018)	V	v	V
Shirazi et al. (2022)	V	V	V
Shollo and Galliers (2016)	V		V
Shollo et al. (2022)	V	v	V
Someh et al. (2019)	V		V
Song et al. (2018)	V	V	V
Song et al. (2022)	V	v	V
Sun et al. (2019)		V	V
Suoniemi et al. (2020)	V		V
Tamm et al. (2020)	V	V	V
Tan et al. (2016)	V	v	V
Tarn and Wang (2023)	V	v	V
Torres et al. (2018)	V		V
Trieu (2017)	V	V	V
Trieu (2023)	V	V	V
Trkman et al. (2010)	V	V	V
Wang et al. (2018)	V	V	V
Wang et al. (2019)	V	v	V
Winig (2017)	V	v	V
Wu and Chang (2012)		V	V
Wu et al. (2020)	V	v	V
Xie et al. (2016)	V	V	V
Yu et al. (2014)	V	V	
Zamani et al. (2021)	V	V	
Zhang et al. (2022)	V		V

* Uppercase V denotes a concept is explicitly presented in literature. Lowercase v denotes a concept is implicitly presented in literature.

Appendix 2. Classification framework and supporting evidence

Construct	Dimension	Items	BDA-specific	Non-BDA-specific
BDA System Capabilities	BDA System Quality	System quality, Big data quality, Information quality	Adya and Phillips-Wren (2020); Ain et al. (2019); Audzeyeva and Hudson (2016); Aversa et al. (2018); Bahrami and Shokouhyar (2022); Bowers et al. (2017); Chatterjee et al. (2023); Constantiou and Kallinikos (2015); Côte-Real et al. (2020); Deng and Chi (2012); Dong and Yang (2020); Elia et al. (2022); Fantini and Narayandas (2023); Fink et al. (2017); Ghasemaghaei (2019b); Ghasemaghaei and Calic (2019); Ghasemaghaei et al. (2018); Ghasemaghaei et al. (2017); Ghasemaghaei and Turel (2022); Grover et al. (2018); Harris and McDonald (2014); Horita et al. (2023); Jha et al. (2020); Kowalczyk and Buxmann (2015); Lakshmi and Corbett (2023); Lebovitz et al. (2022); Marabelli et al. (2021); Mikalef and Krogstie (2020); Mikalef et al. (2020b); Namvar et al. (2021); Naseer et al. (2021); Nichols (2014); Oesterreich et al. (2022); Otondo (2019); Papagiannidis et al. (2023); Popović et al. (2014); Rana et al. (2022); Ross et al. (2013); Saenz et al. (2022); Seddon et al. (2017); Shamim et al. (2019); Shi et al. (2024); Shields (2018); Shollo and Galliers (2016); Someh et al.	Barki et al. (2007); Burton-Jones and Grange (2013); Burton-Jones and Volkoff (2017); Kettinger et al. (2013); O'Connor and O'Reilly (2018)

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Construct	Dimension	Items	BDA-specific	Non-BDA-specific
	BDA System Features and Affordances	BDA materiality, BDA affordances	(2019); Song et al. (2022); Tamm et al. (2020); Tan et al. (2016); Torres et al. (2018); Trieu (2017); Winig (2017); Zhang et al. (2022)	Benlian (2015); Burton-Jones and Volkoff (2017); Pang and Pavlou (2019)
	BDA Strategic Characteristics	Valuable, Rare, Costly to imitate, Organisationally Embedded	Ain et al. (2019); Arnott et al. (2017); Aversa et al. (2018); Bayraktar et al. (2023); Bowers et al. (2017); Chae (2014); Constantiou and Kallinikos (2015); Côte-Real et al. (2020); Deng and Chi (2012); Fantini and Narayandas (2023); Fink et al. (2017); Ganju et al. (2022); İşik et al. (2013); Joshi et al. (2021); Kowalczyk and Buxmann (2015); Lakshmi and Corbett (2023); Lehrer et al. (2018); Mäntymäki et al. (2020); Park et al. (2017); Ransbotham and Kiron (2017); Shao et al. (2023); Shi et al. (2024); Shields (2018); Shollo et al. (2022); Tan et al. (2016); Tarn and Wang (2023); Wang et al. (2018); Winig (2017); Wu et al. (2020); Zhang et al. (2022)	
	BDA Assets	Big data assets (e.g., internal and external), BDA system assets	Elia et al. (2022); Grover et al. (2018); Gupta and George (2016); Mikalef et al. (2020b); Suoniemi et al. (2020)	Burton-Jones and Grange (2013); Kang et al. (2012); Kettinger et al. (2013); Lauterbach et al. (2020)
			Adjerid et al. (2023); Ain et al. (2019); Arnott et al. (2017); Bayraktar et al. (2023); Bowers et al. (2017); Brady et al. (2017); Chae (2014); Chatterjee et al. (2023); Chatterjee et al. (2021); Chen et al. (2015); Chen et al. (2012); Constantiou and Kallinikos (2015); Côte-Real et al. (2020); Elia et al. (2022); Enholm et al. (2022); Fantini and Narayandas (2023); Fink et al. (2017); Ganju et al. (2022); Ghasemaghaei (2019b); Ghasemaghaei et al. (2018); Ghasemaghaei and Turel (2021); Goyal et al. (2012); Grover et al. (2018); Günther et al. (2017); Gupta and George (2016); Harris and McDonald (2014); Horita et al. (2023); Huang et al. (2020); Jha et al. (2020); Joshi et al. (2021); Lakshmi and Corbett (2023); Li et al. (2013); Lin et al. (2022); Mandal (2019); Mäntymäki et al. (2020); Marabelli et al. (2021); Massa and Testa (2005); Mikalef et al. (2020a); Mikalef and Krogstie (2020); Mikalef et al. (2020b); Müller et al. (2018); Namvar et al. (2021); Naseer et al. (2021); Nichols (2014); Oesterreich et al. (2022); Papagiannidis et al. (2023); Park et al. (2017); Popović et al. (2018); Ramakrishnan et al. (2020); Ross et al. (2013); Seddon et al. (2017); Shamim et al. (2019); Shi et al. (2024); Shields (2018); Suoniemi et al. (2020); Tan et al. (2016); Tarn and Wang (2023); Trieu (2017); Wang et al. (2019); Winig (2017); Wu et al. (2020); Xie	

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Construct	Dimension	Items	BDA-specific	Non-BDA-specific
BDA Users and Tasks	User Types	Technical users, Managerial users, Domain experts, External users (e.g., customers, suppliers)	et al. (2016); Zamani et al. (2021); Zhang et al. (2022) Arnott et al. (2017); Audzeyeva and Hudson (2016); Aversa et al. (2018); Brady et al. (2017); Deng and Chi (2012); Elia et al. (2022); Işık et al. (2013); Joshi et al. (2021); Kowalczyk and Buxmann (2015); Massa and Testa (2005); Mikalef et al. (2020b); Oesterreich et al. (2022); Seddon et al. (2017); Shamim et al. (2019); Shao et al. (2023); Shollo et al. (2022); Tamm et al. (2020); Tan et al. (2016); Torres et al. (2018); Trieu (2017); Yu et al. (2014)	
	User Expertise	User knowledge, User skills, User experience	Ain et al. (2019); Arnott et al. (2017); Audzeyeva and Hudson (2016); Aversa et al. (2018); Bowers et al. (2017); Brady et al. (2017); Chen et al. (2022); Constantiou and Kallinikos (2015); Deng and Chi (2012); Ebrahimi and Matt (2023); Elia et al. (2022); Fantini and Narayandas (2023); Ghasemaghaei (2019b); Ghasemaghaei et al. (2018); Ghasemaghaei et al. (2017); Ghasemaghaei and Turel (2022); Goyal et al. (2012); Grover et al. (2018); Günther et al. (2017); Gupta and George (2016); Horita et al. (2023); Joshi et al. (2021); Kowalczyk and Buxmann (2015); Lebovitz et al. (2022); Li et al. (2013); Mikalef et al. (2020a); Mikalef and Krogstie (2020); Mikalef et al. (2020b); Namvar et al. (2021); Oesterreich et al. (2022); Otondo (2019); Papagiannidis et al. (2023); Park et al. (2017); Popović et al. (2018); Ross et al. (2013); Seddon et al. (2017); Shamim et al. (2019); Shao et al. (2023); Shields (2018); Shollo and Galliers (2016); Someh et al. (2019); Suoniemi et al. (2020); Tamm et al. (2020); Tan et al. (2016); Torres et al. (2018); Trieu (2017); Winig (2017); Yu et al. (2014)	Benlian (2015); Burton-Jones and Grange (2013); Kohli and Melville (2019); O'Connor and O'Reilly (2018); Sun et al. (2019)
	User Motivation	Intrinsic motivation, Extrinsic motivation, Attitudes, Beliefs	Adya and Phillips-Wren (2020); Brady et al. (2017); Ebrahimi and Matt (2023); Ghasemaghaei and Turel (2022); Lakshmi and Corbett (2023); Lebovitz et al. (2022); Li et al. (2013); Namvar et al. (2021); Popović et al. (2014); Winig (2017)	Barki et al. (2007); Beaudry et al. (2020); Benlian (2015); Burton-Jones and Grange (2013); Kang et al. (2012); Kohli and Melville (2019); O'Connor and O'Reilly (2018); Sun et al. (2019)
	User Goals	User objectives, User goal type (e.g., learning goals & outcome goals)	Ghasemaghaei and Turel (2022); Lehrer et al. (2018)	Burton-Jones and Grange (2013); Burton-Jones and Volkoff (2017)
	Task Structure	Deep/Surface structure, Structured/Semi-structured/ Unstructured task (e.g., operational, tactical, strategic), Task discretion	Adjerid et al. (2023); Arnott et al. (2017); Côte-Real et al. (2020); Işık et al. (2013); Kowalczyk and Buxmann (2015); Lebovitz et al. (2022); Shollo et al. (2022); Trieu (2023)	Burton-Jones and Grange (2013); Kuegler et al. (2015); Lauterbach et al. (2020); O'Connor and O'Reilly (2018)
	Task Complexity	Task-system co-dependency, Task analysability, Task variation, Data	Adjerid et al. (2023); Arnott et al. (2017); Aversa et al. (2018); Côte-Real et al. (2020); Ebrahimi and	Burton-Jones and Grange (2013); Lauterbach et al. (2020)

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Construct	Dimension	Items	BDA-specific	Non-BDA-specific
BDA Use	Routine Use	complexity, Computational complexity, System complexity	Matt (2023); Ganju et al. (2022); Ghasemaghahi and Turel (2022); Lebovitz et al. (2022); Trieu (2023)	
		Frequency of BDA use, Duration of BDA use	Ain et al. (2019); Arnott et al. (2017); Audzeyeva and Hudson (2016); Bahrami and Shokouhyar (2022); Bowers et al. (2017); Božić and Dimovski (2019); Brady et al. (2017); Chen et al. (2015); Chen et al. (2021); Constantiou and Kallinikos (2015); Deng and Chi (2012); Dong and Yang (2020); Drydakis (2022); Enholm et al. (2022); Fantini and Narayandas (2023); Fink et al. (2017); Ghasemaghahi (2019b); Ghasemaghahi and Calic (2019); Ghasemaghahi et al. (2017); Goyal et al. (2012); Grover et al. (2018); Günther et al. (2017); Horita et al. (2023); Işık et al. (2013); Jha et al. (2020); Lakshmi and Corbett (2023); Lebovitz et al. (2022); Lehrer et al. (2018); Li et al. (2013); Lin et al. (2022); Marabelli et al. (2021); Massa and Testa (2005); Nichols (2014); Otondo (2019); Papagiannidis et al. (2023); Saldanha et al. (2017); Seddon et al. (2017); Shamim et al. (2019); Shi et al. (2022); Shi et al. (2024); Shollo and Galliers (2016); Shollo et al. (2022); Someh et al. (2019); Song et al. (2018); Tan et al. (2016); Tarn and Wang (2023); Trieu (2017); Trkman et al. (2010); Wang et al. (2019); Winig (2017); Wu et al. (2020); Xie et al. (2016); Yu et al. (2014)	Alalwan et al. (2014); Barki et al. (2007); Beaudry et al. (2020); Benlian (2015); Chen et al. (2020); Dwivedi et al. (2023); Kang et al. (2012); Kuegler et al. (2015); Li et al. (2019); O'Connor and O'Reilly (2018); Sun et al. (2019); Wu and Chang (2012)
		Revising: content of features in use; spirit of features in use	Deng and Chi (2012); Joshi et al. (2021); Li et al. (2013); Ransbotham and Kiron (2017)	Beaudry et al. (2020); Kohli and Melville (2019); O'Connor and O'Reilly (2018); Sun et al. (2019)
		Operational adaptation, Organisational adaptation, Technology adaptation	Chae (2014); Deng and Chi (2012)	Barki et al. (2007); Beaudry et al. (2020)
Goal-oriented Activities	Decision-related Activities	Effective use of BDA, Effective use of information	Adya and Phillips-Wren (2020); Ain et al. (2019); Aversa et al. (2018); Bao and Bardhan (2022); Ebrahimi and Matt (2023); Ganju et al. (2022); Ghasemaghahi and Turel (2022); Harris and McDonald (2014); Joshi et al. (2021); Kowalczyk and Buxmann (2015); Lebovitz et al. (2022); Mäntymäki et al. (2020); Massa and Testa (2005); Namvar et al. (2021); Naseer et al. (2021); Popović et al. (2014); Ross et al. (2013); Shields (2018); Shirazi et al. (2022); Song et al. (2022); Tamm et al. (2020); Trieu (2017, 2023); Wang et al. (2018); Zamani et al. (2021)	Beaudry et al. (2020); Burton-Jones and Grange (2013); Burton-Jones and Volkoff (2017); Chatterjee et al. (2020); Chen et al. (2020); Kettinger et al. (2013); Lauterbach et al. (2020); O'Connor and O'Reilly (2018); Pang and Pavlou (2019)
		Using BDA insights, Evaluating alternative courses of actions, Choosing these alternatives to proceed	Ain et al. (2019); Audzeyeva and Hudson (2016); Aversa et al. (2018); Chen et al. (2015); Chen et al. (2021); Constantiou and Kallinikos (2015); Fantini and Narayandas (2023); Harris and McDonald (2014); Kowalczyk and	Alalwan et al. (2014); Burton-Jones and Volkoff (2017)

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Construct	Dimension	Items	BDA-specific	Non-BDA-specific
Intermediate Outcomes	Task-related Activities	Taking; informed actions; competitive actions	Buxmann (2015); Lebovitz et al. (2022); Massa and Testa (2005); Naseer et al. (2021); Otondo (2019); Ramakrishnan et al. (2020); Rana et al. (2022); Ross et al. (2013); Saenz et al. (2022); Seddon et al. (2017); Shamim et al. (2019); Shi et al. (2024); Shields (2018); Shollo and Galliers (2016); Shollo et al. (2022); Someh et al. (2019); Torres et al. (2018); Trieu (2017); Winig (2017); Wu et al. (2020); Zamani et al. (2021) Ain et al. (2019); Bahrami and Shokouhyar (2022); Bowers et al. (2017); Ghasemaghaei and Turel (2021); Harris and McDonald (2014); Lebovitz et al. (2022); Namvar et al. (2021); Naseer et al. (2021); Saldanha et al. (2017); Seddon et al. (2017); Shamim et al. (2019); Shi et al. (2024); Shields (2018); Shirazi et al. (2022); Shollo et al. (2022); Torres et al. (2018); Wang et al. (2018); Wu et al. (2020); Zamani et al. (2021)	Benlian (2015); Burton-Jones and Grange (2013); Burton-Jones and Volkoff (2017); Chen et al. (2020); Kettinger et al. (2013); O'Connor and O'Reilly (2018); Sun et al. (2019)
	Decision-related Outcomes	Insight generation, Support of decision-making processes, Improved decision effectiveness and efficiency	Adya and Phillips-Wren (2020); Ain et al. (2019); Audzeyeva and Hudson (2016); Aversa et al. (2018); Bayraktar et al. (2023); Chae (2014); Chatterjee et al. (2021); Chen et al. (2021); Chen et al. (2022); Constantiou and Kallinikos (2015); Drydakis (2022); Enholm et al. (2022); Fantini and Narayandas (2023); Ghasemaghaei (2019b); Ghasemaghaei and Calic (2019); Ghasemaghaei et al. (2018); Ghasemaghaei et al. (2017); Ghasemaghaei and Turel (2021); Goyal et al. (2012); Grover et al. (2018); Horita et al. (2023); Kowalczyk and Buxmann (2015); Lin et al. (2022); Mandal (2019); Naseer et al. (2021); Otondo (2019); Park et al. (2017); Ramakrishnan et al. (2020); Rana et al. (2022); Saenz et al. (2022); Seddon et al. (2017); Shamim et al. (2019); Shields (2018); Shollo and Galliers (2016); Shollo et al. (2022); Someh et al. (2019); Song et al. (2018); Suoniemi et al. (2020); Tan et al. (2016); Trieu (2017, 2023); Trkman et al. (2010); Wang et al. (2018); Zhang et al. (2022)	Alalwan et al. (2014); Beaudry et al. (2020); Burton-Jones and Volkoff (2017); Chatterjee et al. (2020); Kuegler et al. (2015); Wu and Chang (2012)
	Task-related Outcomes	Improved task performance: effectiveness; efficiency; innovation; productivity	Ain et al. (2019); Bowers et al. (2017); Drydakis (2022); Enholm et al. (2022); Ghasemaghaei et al. (2017); Joshi et al. (2021); Lakshmi and Corbett (2023); Lin et al. (2022); Mandal (2019); Naseer et al. (2021); Saenz et al. (2022); Shao et al. (2023); Song et al. (2022); Song et al. (2018); Suoniemi et al. (2020); Tamm et al. (2020); Tan et al. (2016); Tarn and Wang (2023); Trkman et al. (2010); Wang et al. (2018); Zhang et al. (2022)	Barki et al. (2007); Beaudry et al. (2020); Benlian (2015); Burton-Jones and Grange (2013); Burton-Jones and Volkoff (2017); Chatterjee et al. (2020); Chen et al. (2020); Dwivedi et al. (2023); Kang et al. (2012); Kettinger et al. (2013); Kohli and Melville (2019); Kuegler et al. (2015); Lauterbach et al. (2020); O'Connor and O'Reilly (2018); Sun et al. (2019)

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Construct	Dimension	Items	BDA-specific	Non-BDA-specific
Ultimate Outcomes	Operational Value	Business operational efficiency, Business process improvement, Increase in firm productivity	Audzeyeva and Hudson (2016); Bayraktar et al. (2023); Chatterjee et al. (2023); Chatterjee et al. (2021); Chen et al. (2015); Drydakis (2022); Elia et al. (2022); Enholm et al. (2022); Fink et al. (2017); Ghasemaghaei et al. (2017); Grover et al. (2018); Gupta and George (2016); Huang et al. (2020); Mikalef et al. (2020b); Müller et al. (2018); Oesterreich et al. (2022); Popović et al. (2018); Raguseo et al. (2021); Rana et al. (2022); Ross et al. (2013); Seddon et al. (2017); Someh et al. (2019); Song et al. (2018); Torres et al. (2018); Trieu (2017); Wang et al. (2018); Wu et al. (2020); Xie et al. (2016)	Burton-Jones and Volkoff (2017); Li et al. (2019); Wu and Chang (2012)
	Social Value	Improved social wellbeing, Improved sustainability	Adjerid et al. (2023); Bao and Bardhan (2022); Chatterjee et al. (2023); Drydakis (2022); Enholm et al. (2022); Ganju et al. (2022); Günther et al. (2017); Lakshmi and Corbett (2023); Shi et al. (2024); Xie et al. (2016)	Pang and Pavlou (2019)
	Economic Value	Profit increase, Business growth, Increase in market share	Aversa et al. (2018); Bahrami and Shokouhyar (2022); Bayraktar et al. (2023); Božić and Dimovski (2019); Chatterjee et al. (2023); Chatterjee et al. (2021); Côte-Real et al. (2020); Dong and Yang (2020); Drydakis (2022); Enholm et al. (2022); Ganju et al. (2022); Goyal et al. (2012); Günther et al. (2017); Huang et al. (2020); Nichols (2014); Oesterreich et al. (2022); Papagiannidis et al. (2023); Ramakrishnan et al. (2020); Rana et al. (2022); Saenz et al. (2022); Shi et al. (2022); Shields (2018); Shirazi et al. (2022); Song et al. (2022); Suoniemi et al. (2020); Torres et al. (2018); Xie et al. (2016)	Li et al. (2019); Wu and Chang (2012)
	Organisational Value	Improved BDA capabilities, Improvement of organisational knowing	Ain et al. (2019); Audzeyeva and Hudson (2016); Bao and Bardhan (2022); Božić and Dimovski (2019); Chen et al. (2012); Constantiou and Kallinikos (2015); Elia et al. (2022); Goyal et al. (2012); Harris and McDonald (2014); Jha et al. (2020); Lin et al. (2022); Mikalef et al. (2020b); Nichols (2014); Oesterreich et al. (2022); Papagiannidis et al. (2023); Park et al. (2017); Seddon et al. (2017); Shollo and Galliers (2016); Shollo et al. (2022); Tamm et al. (2020); Trieu (2017); Wang et al. (2018); Winig (2017); Wu et al. (2020); Xie et al. (2016); Zhang et al. (2022)	Wu and Chang (2012)
	Strategic Value	Gaining and sustaining competitive advantage, Product, and service innovation	Aversa et al. (2018); Božić and Dimovski (2019); Chae (2014); Chatterjee et al. (2023); Chatterjee et al. (2021); Chen et al. (2015); Chen et al. (2012); Constantiou and Kallinikos (2015); Côte-Real et al. (2020); Elia et al. (2022); Enholm et al. (2022); Fink et al. (2017);	Kohli and Melville (2019); Li et al. (2019)

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Construct	Dimension	Items	BDA-specific	Non-BDA-specific
BDA Organisational Capabilities	Organisational Analytics Capability Development	Establishing BDA infrastructure, Ability to: pre-process, integrate and disseminate data; handle data volume/variety; design analysis; interpret outputs, Data governance	Grover et al. (2018); Gupta and George (2016); Jha et al. (2020); Lehrer et al. (2018); Mikalef et al. (2020a); Mikalef and Krogstie (2020); Mikalef et al. (2020b); Oesterreich et al. (2022); Papagiannidis et al. (2023); Raguseo et al. (2021); Rana et al. (2022); Ransbotham and Kiron (2017); Ross et al. (2013); Saldanha et al. (2017); Shirazi et al. (2022); Someh et al. (2019); Trieu (2017); Wang et al. (2019); Wang et al. (2018); Wu et al. (2020)	Kettinger et al. (2013)
	Organisational Readiness	Financial capital, IT sophistication	Ain et al. (2019); Audzeyeva and Hudson (2016); Aversa et al. (2018); Bao and Bardhan (2022); Bowers et al. (2017); Božić and Dimovski (2019); Chen et al. (2021); Chen et al. (2022); Constantiou and Kallinikos (2015); Elia et al. (2022); Enholm et al. (2022); Fink et al. (2017); Ghasemaghaei (2019b); Goyal et al. (2012); Grover et al. (2018); Günther et al. (2017); Harris and McDonald (2014); Huang et al. (2020); Jha et al. (2020); Joshi et al. (2021); Kowalczyk and Buxmann (2015); Mandal (2019); Marabelli et al. (2021); Massa and Testa (2005); Mikalef et al. (2020a); Mikalef et al. (2020b); Nichols (2014); Oesterreich et al. (2022); Papagiannidis et al. (2023); Popović et al. (2018); Raguseo et al. (2021); Ramakrishnan et al. (2020); Rana et al. (2022); Ransbotham and Kiron (2017); Ross et al. (2013); Seddon et al. (2017); Shamim et al. (2019); Shi et al. (2022); Shields (2018); Shollo et al. (2022); Someh et al. (2019); Suoniemi et al. (2020); Tan et al. (2016); Torres et al. (2018); Trieu (2017); Wang et al. (2019); Winig (2017); Xie et al. (2016); Zhang et al. (2022)	
	Data-driven Culture	Data-driven, evidence-based decision-making embedded in an organisation's DNA, Analytics leadership	Chen et al. (2015); Côte-Real et al. (2020); Elia et al. (2022); Enholm et al. (2022); Gupta and George (2016); Jha et al. (2020); Mandal (2019); Mikalef et al. (2020a); Mikalef and Krogstie (2020); Mikalef et al. (2020b); Oesterreich et al. (2022); Popović et al. (2018); Raguseo et al. (2021); Zhang et al. (2022)	
			Bowers et al. (2017); Elia et al. (2022); Fantini and Narayandas (2023); Goyal et al. (2012); Grover et al. (2018); Gupta and George (2016); Mandal (2019); Marabelli et al. (2021); Mikalef et al. (2020a); Mikalef and Krogstie (2020); Mikalef et al. (2020b); Nichols (2014); Oesterreich et al. (2022); Otondo (2019); Raguseo et al. (2021); Ramakrishnan et al. (2020); Ransbotham and Kiron (2017); Ross et al. (2013); Seddon et al. (2017);	Kettinger et al. (2013)

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Construct	Dimension	Items	BDA-specific	Non-BDA-specific
Complementary Organisational Resources	Organisational Deep Structure	Organisational culture, Structure, Power distribution, Management control, Absorptive capacity	Shamim et al. (2019); Shao et al. (2023); Shi et al. (2022); Shi et al. (2024); Shields (2018); Shollo and Galliers (2016); Someh et al. (2019); Suoniemi et al. (2020); Trieu (2023); Winig (2017) Adjerid et al. (2023); Ain et al. (2019); Audzeyeva and Hudson (2016); Aversa et al. (2018); Bahrami and Shokouhyar (2022); Bao and Bardhan (2022); Bowers et al. (2017); Brady et al. (2017); Chen et al. (2015); Chen et al. (2022); Constantiou and Kallinikos (2015); Côte-Real et al. (2020); Elia et al. (2022); Enholm et al. (2022); Fantini and Narayandas (2023); Goyal et al. (2012); Grover et al. (2018); Günther et al. (2017); Harris and McDonald (2014); Horita et al. (2023); Joshi et al. (2021); Kowalczyk and Buxmann (2015); Lakshmi and Corbett (2023); Mäntymäki et al. (2020); Mikalef and Krogstie (2020); Mikalef et al. (2020b); Nichols (2014); Otondo (2019); Park et al. (2017); Popović et al. (2018); Ross et al. (2013); Saldanha et al. (2017); Seddon et al. (2017); Shi et al. (2022); Shi et al. (2024); Shields (2018); Shollo et al. (2022); Trieu (2017); Trkman et al. (2010); Wang et al. (2019); Winig (2017); Zamani et al. (2021)	Chatterjee et al. (2020); Kohli and Melville (2019)
	Organisation Strategy	Organisational goals, Exploitation strategies, Exploration strategies	Ain et al. (2019); Audzeyeva and Hudson (2016); Aversa et al. (2018); Constantiou and Kallinikos (2015); Fantini and Narayandas (2023); Fink et al. (2017); Goyal et al. (2012); Grover et al. (2018); Günther et al. (2017); Joshi et al. (2021); Lakshmi and Corbett (2023); Lehrer et al. (2018); Marabelli et al. (2021); Mikalef and Krogstie (2020); Otondo (2019); Ross et al. (2013); Shi et al. (2024); Shields (2018); Shollo et al. (2022); Someh et al. (2019); Trieu (2017); Winig (2017); Zhang et al. (2022)	Beaudry et al. (2020)
	Organisation Knowledge	Domain knowledge (e.g., marketing)	Bowers et al. (2017); Harris and McDonald (2014); Mikalef et al. (2020b); Shi et al. (2024); Shollo et al. (2022); Tarn and Wang (2023)	
	Technological Resources	Information systems and data contained (e.g., customer relationship management)	Chen et al. (2022); Enholm et al. (2022); Mikalef et al. (2020b); Saldanha et al. (2017); Someh et al. (2019); Trkman et al. (2010); Xie et al. (2016)	
	Non-technological Resources	Human resources and financial resources	Bao and Bardhan (2022); Goyal et al. (2012); Lakshmi and Corbett (2023); Lebovitz et al. (2022); Popović et al. (2018); Ross et al. (2013); Shields (2018); Winig (2017)	Kohli and Melville (2019)
BDA Learning	Individual Adaptation Behaviours	Learning to effective use BDA, Communication, Independent exploration, IT feature broadening, IT feature deepening, Feedback	Chae (2014); Deng and Chi (2012); Li et al. (2013); Massa and Testa (2005); Namvar et al. (2021); Seddon et al. (2017); Zamani et al. (2021)	Barki et al. (2007); Benlian (2015); Burton-Jones and Grange (2013); Burton-Jones and Volkoff (2017); Lauterbach et al. (2020)

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Construct	Dimension	Items	BDA-specific	Non-BDA-specific
	Organisational Learning	Negotiations, Articulation, Contestations, Data selection, Training	Bowers et al. (2017); Chae (2014); Grover et al. (2018); Gupta and George (2016); Jha et al. (2020); Li et al. (2013); Otondo (2019); Papagiannidis et al. (2023); Ross et al. (2013); Seddon et al. (2017); Shields (2018); Shollo and Galliers (2016); Trieu (2017)	
Internal and External Environment	Internal – Organisational Characteristics	Organisational inertia, Organisation size	Adjerid et al. (2023); Audzeyeva and Hudson (2016); Božić and Dimovski (2019); Dong and Yang (2020); Mikalef and Krogstie (2020); Park et al. (2017); Seddon et al. (2017); Suoniemi et al. (2020); Trieu (2017)	Kohli and Melville (2019); Wu and Chang (2012)
	External – Industry Characteristics	External pressure, Dynamism, Competitive position, Sector, IT intensity, Heterogeneity, Competition	Adjerid et al. (2023); Aversa et al. (2018); Bayraktar et al. (2023); Božić and Dimovski (2019); Brady et al. (2017); Chae (2014); Chen et al. (2015); Chen et al. (2022); Constantiou and Kallinikos (2015); Côte-Real et al. (2020); Grover et al. (2018); Jha et al. (2020); Mäntymäki et al. (2020); Mikalef et al. (2020a); Mikalef and Krogstie (2020); Müller et al. (2018); Naseer et al. (2021); Oesterreich et al. (2022); Park et al. (2017); Shi et al. (2024); Shirazi et al. (2022); Song et al. (2022); Song et al. (2018); Suoniemi et al. (2020); Trieu (2017); Winig (2017); Zamani et al. (2021)	Kettinger et al. (2013); Kohli and Melville (2019); Wu and Chang (2012)
	External – Country Characteristics	Law, Infrastructure, Culture	Ganju et al. (2022); Lakshmi and Corbett (2023); Trieu (2017); Wang et al. (2019)	

Data availability

Data will be made available on request.

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